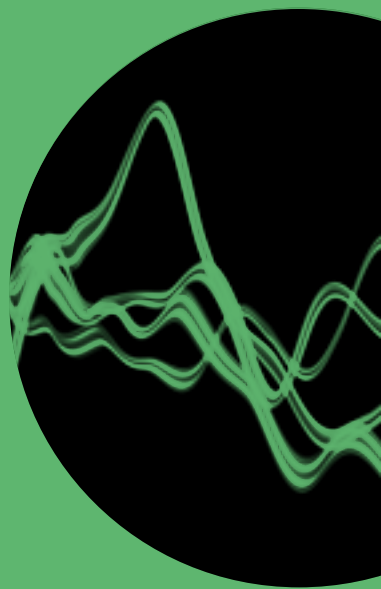


Review

ERICSSON
TECHNOLOGY



EVOLVING
NETWORKS FOR
**EXTENDED
REALITY**

THREE TECH
TRENDS DRIVING
**NEXT-LEVEL
DIGITALIZATION**

3GPP TECHNOLOGY
FOR **SATELLITE
COMMUNICATION**







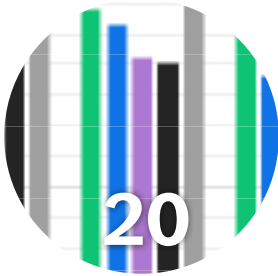
08 FUTURE NETWORK REQUIREMENTS FOR EXTENDED REALITY APPLICATIONS

The rapid development of extended reality (XR) technologies and devices has major implications for mobile networks. Communication service providers need enhanced network capabilities to deliver the low latency, high reliability and high data rates that are essential to sustain user quality of experience in XR applications.



20 NETWORK EVOLUTION TO SUPPORT EXTENDED REALITY APPLICATIONS

To meet the stringent requirements of XR applications on latency, reliability and bitrates, mobile networks must evolve to include time-critical communication features, and many deployments will need to be densified. Our research indicates that delay-aware scheduling and robust link adaptation are effective techniques to meet XR requirements in the near term.



30 FEATURE ARTICLE

Networking trends 2023: Building the platform for next-level digitalization

Ericsson's CTO explains how high-performance networks, enterprise networking and network application programming interfaces for monetization will help to create an open business innovation platform that connects consumers, enterprises and society in a cyber-physical world without boundaries.



42 USING 3GPP TECHNOLOGY FOR SATELLITE COMMUNICATION

The most efficient way to provide truly global mobile coverage using everyday user equipment such as smartphones is to integrate a satellite component into 5G Advanced and 6G mobile network systems that is based on open 3GPP (3rd Generation Partnership Project) specifications.



54 AUTONOMOUS NETWORKS WITH MULTI-LAYER, INTENT-BASED OPERATION

Intent-management solutions can deliver a high level of autonomy in network operation. With the help of techniques including digital-twin-based utility evaluation and machine-learning-based automated exploration of system behavior, Ericsson researchers have created a highly self-adaptive and reliable autonomous network with minimal need for human involvement.



Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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EXTENDED REALITY, SATELLITES AND NEXT-LEVEL DIGITALIZATION

■ **THE ONGOING DIGITALIZATION** occurring all around the globe is profoundly changing the way both individuals and organizations consume products and services. As impressive as this transformation has been, I believe that what we have seen so far is just the beginning. The emergence of increasingly sophisticated cloud and artificial intelligence technologies, programmable networks, application programming interface exposure and as-a-service business models all point toward the rapidly growing need for an open platform for business innovation.

At Ericsson, we are confident that 5G will evolve into the network infrastructure of choice. There is simply no other communication technology that can efficiently deliver the required openness, scale and affordability. Based on this conviction, we are working closely with our standardization partners to create a truly open and business-innovation-friendly network platform that links consumers, enterprises and society at large with seamless and efficient connectivity and differentiated communication services. By minimizing both cost and resource usage, the future network platform will also play a key role in helping society and industry to reach net-zero emissions and other sustainability goals.

If you are curious to know more about my vision for the future of digitalization, check out my annual technology trends article on page 30 of this issue. In it, I present the three key trends for 2023 and chart the course toward

WE ARE CONFIDENT THAT 5G WILL EVOLVE INTO THE NETWORK INFRASTRUCTURE OF CHOICE

achieving next-level digitalization – that is, the creation of a connected and sustainable physical world that is both digitalized and programmable.

This issue of our magazine also contains four other articles that are closely connected to the theme of next-level digitalization. Two of them focus on the critically important topic of extended reality (XR). The first one, on page 8, explores the network requirements of the most likely future XR applications and defines a set of XR flavors to serve as samples in the expected requirements range. In the second XR article, on page 20, the same team of experts examines the traffic characteristics of XR applications and explains the qualitative effects they have on both coverage and capacity.

Aside from XR support, next-level digitalization also requires ubiquitous connectivity, which is where satellite connectivity comes in. The article on page 42 provides an overview of satellite technology and explains the use cases, service requirements and business rationales for integrating a satellite component based on the 3GPP ecosystem into 5G Advanced and 6G mobile network systems.

Of course, next-level digitalization also requires a high degree of autonomous operation, which is best achieved through the use of intents. In the article on page 52, a team of Ericsson experts explains how to use intent-based interfaces to

achieve autonomous operation, addressing key concepts such as decision-making based on utility maximization and the use of digital twins for safe, automated exploration of solution alternatives.

I hope that you enjoy this issue of Ericsson Technology Review and find it useful in your work. Please feel free to share it with your colleagues and business partners. You can find both PDF and HTML versions of all the articles at: ericsson.com/ericsson-technology-review



ERIK EKUDDEN
SENIOR VICE PRESIDENT,
CHIEF TECHNOLOGY OFFICER

Future network requirements for extended reality applications

Many experts believe that new and emerging extended reality technologies will lead to the next major paradigm shift in telecommunications, with lightweight XR glasses ultimately overtaking smartphones as the dominant device type in mobile networks. This evolution has major implications on the requirements for future networks.

FREDRIK ALRIKSSON,
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YING SUN

The availability of discrete, attractive and high-performing extended reality (XR) devices is key to the mass-market success of XR applications, and the process of creating them is well underway. Device developers are working intensively to evolve the device form factor from today’s bulky head-mounted displays (HMDs) to lightweight and stylish glasses that will depend on computation offloading to a much greater extent.

■ It is of the utmost importance that future networks are designed and dimensioned to meet the requirements of XR applications. Low latency, high reliability and high data rates will be essential to sustain user quality of experience (QoE) in XR applications while offloading computation services. XR applications are also going to generate a significant amount of additional traffic, which networks must have the capacity to handle. To gain a better understanding of XR network

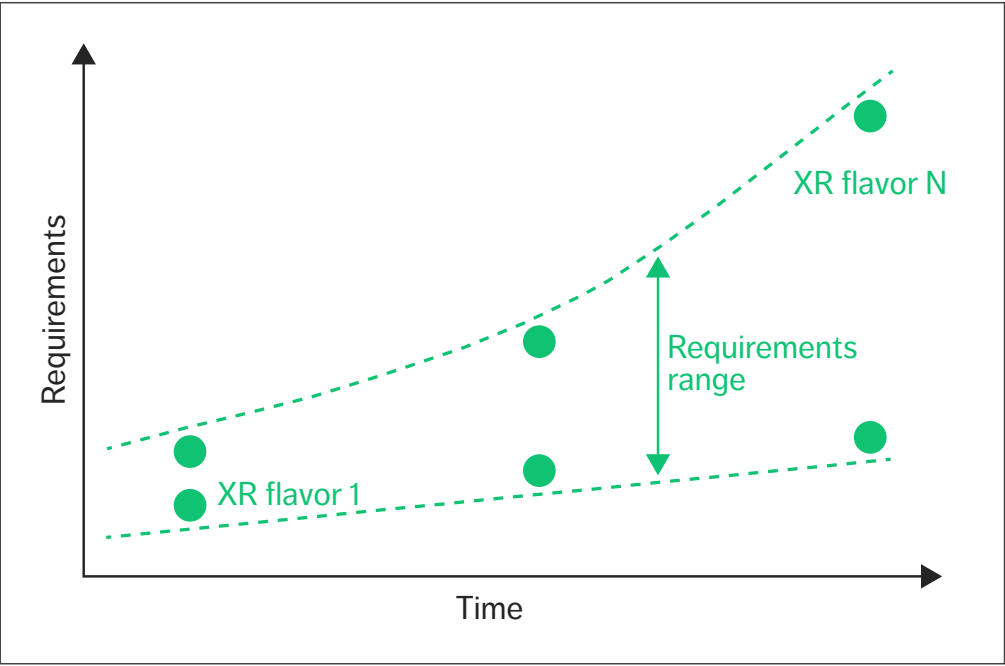


Figure 1 XR requirements with stringency increasing over time

requirements, Ericsson has derived a set of XR operating points as concrete samples from the requirements space, as illustrated in Figure 1. For the sake of simplicity, we refer to the selected samples (represented by the dots in Figure 1) as XR “flavors.” We have used the flavors to make projections regarding how the network requirements of XR applications, devices and computation offloading variants are likely to evolve over time, and what implications the evolving requirements will have in terms of network capabilities. To start, we identified which XR applications are

most likely to be used and how advanced we expect them to be. In the next step, we considered computation offloading and its impact on connectivity. The XR flavors result from a combination of these two aspects, together with device capabilities, that enables us to derive the requirements on the network for these specific examples.

XR applications and user behavior

Extended reality is an umbrella term that covers immersive technologies ranging from virtual reality

Terms and abbreviations

AR – Augmented Reality | DL – Downlink | HMD – Head-Mounted Display | MBB – Mobile Broadband | Mbps – Megabits per Second | ms – Milliseconds | QoE – Quality of Experience | RAN – Radio-Access Network | SLAM – Simultaneous Localization and Mapping | UL – Uplink | XR – Extended Reality

🕶️ DAILY XR USAGE IS EXPECTED TO INCREASE SIGNIFICANTLY AS XR DEVICES BECOME SLIMMER AND MORE CONVENIENT TO WEAR FOR LONGER PERIODS 🕶️

to mixed reality and augmented reality (AR) [1]. XR devices are expected to bring immersive experiences that merge digital objects, information and overlays with the real world. XR represents three major shifts:

- 1. from flat screens and 2D content to immersive 3D content displayed and spatially located in the real world
- 2. from graphical user interfaces to natural interaction within the real world
- 3. from context switching (between screen and real world) to context awareness.

Application areas that have a high potential for wide uptake include gaming and entertainment, retail and shopping, social communication and virtualized work. These comprise consumption of XR content in a single user setting as well as communication services such as holographic communication that include two or more XR users [2].

XR applications differ in terms of complexity of graphics, as well as on how aligned the graphics need to be to the real environment. The range of XR applications includes everything from a simple navigation service with an arrow pointing toward a destination to advanced holographic communication with moving 3D objects perfectly aligned with the real environment. Unsurprisingly, the most advanced applications with the highest degree of computation offloading have the greatest impact on network requirements, and the simplest ones with the least amount of computation offloading have the smallest impact. It should be noted, however, that a

single application can be elastic in the sense that it can operate with different resolutions, for example. Network requirements are also impacted by usage location and the distribution of use over the time of day. Consumer studies indicate that the introduction of XR may not lead to abrupt changes in user patterns – that is, consumers will likely continue to use the same type of services at the same locations and times as now. In this sense, XR is “only” a new type of user interface. Over time, XR is expected to create additional value for consumers across a broad range of application areas [3, 4].

Daily XR usage is expected to increase significantly as XR devices become slimmer and more convenient to wear for longer periods. But even when it becomes feasible for large numbers of people to wear XR devices all day, it is reasonable to expect the types of XR services that are used to vary over the course of the day, similar to the usage patterns of smartphone services today. Therefore, we expect it would be adequate to dimension future networks to support heavy XR usage for as little as one hour, or just a few hours, per day.

XR functions and computation offloading

Many of the functions needed to realize XR experiences are computationally heavy. Rendering and spatial computation are the two most important. Rendering refers to the process of generating graphics to be shown to the user in the HMD. Depending on the complexity and resolution of the graphics, rendering can become very computationally demanding. Spatial computation functionalities include simultaneous localization and mapping (SLAM), object detection and object tracking. These are primarily used to gain an understanding of the local environment.

Rendering and spatial computation are both good candidates for offloading to a remote server – that is, an edge or cloud server with adequate computation capability. Computation offloading enables more powerful processing through cloud computation and reduced HMD complexity in terms of a smaller form factor, increased battery life and reduced heat generation. Cloud computing allows for more

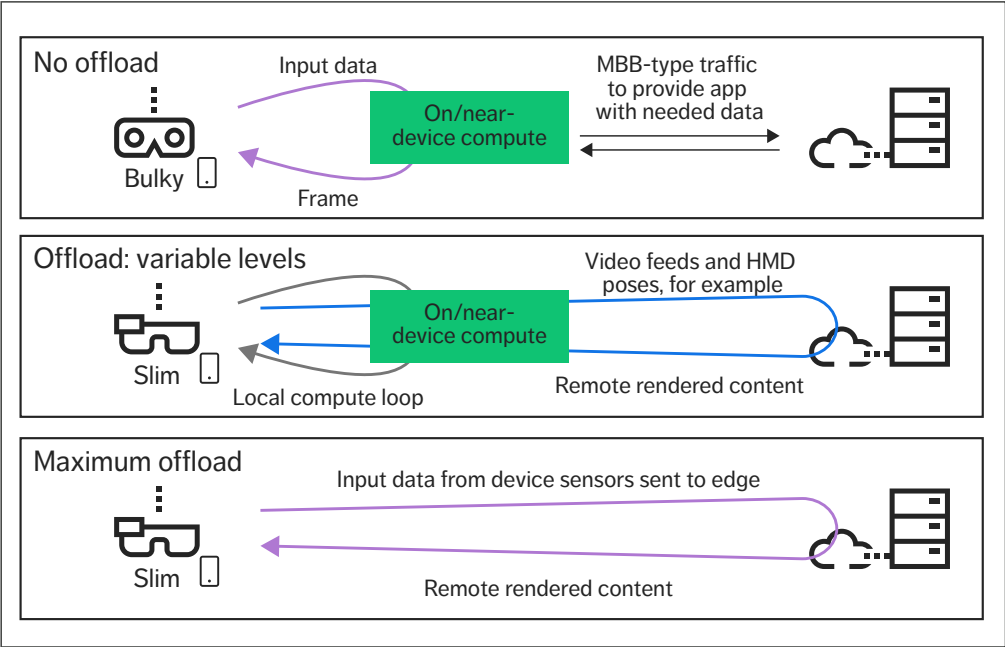


Figure 2 Three approaches to XR computation processing

complex (and faster) computations that enable more immersive experiences with higher resolution, quality and frame rates, as well as supporting multi-user experiences in dynamic environments.

Figure 2 illustrates the three main approaches to XR computation processing. In the “no offload” scenario at the top, all the computation occurs in the device itself or with the support of a tethered companion device nearby (such as a smartphone). In the “maximum offload” scenario at the bottom, all computation occurs remotely, in the cloud. Previously referred to as a “high offload” option [1], maximum offload has extreme requirements on connectivity and is not expected to reach mainstream adoption in the foreseeable future.

In the “offload: variable levels” scenario in the middle, some computation occurs in or near the device, while other computation occurs in the cloud. This split may be user-specific and possibly dynamic during the service period depending on the device

capabilities, cloud capabilities, network functionalities and conditions, and user preferences. This scenario corresponds to the “low offload” and “mid offload” options defined in a previous article [1] and represents the set of offload scenarios we expect will gain market traction.

Offloading as a mechanism implies data transmission between device and the processing unit on the network side, where the resulting bit rate and latency requirements vary depending on many factors.

In the case of remote rendering, the application graphics are rendered at a remote server (an edge server, for example) and sent to the HMD through a 5G (or other) network. Cloud gaming is a contemporary example of this. Remote rendering requires high download (DL) bitrate with bounded low delay to provide good QoE. Typically, two or more 2D video streams are sent in the DL. When foveated rendering is used, four 2D streams of

different resolutions will be sent to the HMD. Another version of remote rendering is to render volumetric content remotely and let the HMD make the final 2D rendering based on the most up-to-date pose data. In this case, HMD-local, latency-hiding techniques such as space warp may use the 3D data streams to enable the user to look at an object from other angles without requiring more data in the DL.

When offloading spatial computation functions, the HMD uploads compressed sensor data (video, lidar and so on) or detected objects to a server that builds a map of the environment and positions the HMD within it. Offloading SLAM enables shared immersive multi-user experiences, as the map of the environment can easily be shared between users in the remote server. These maps can make the positioning task of an HMD that is entering a previously mapped environment more efficient and allow for object persistence. There is a variety of offloading options for SLAM, and complete offload implies very stringent latency requirements. The uplink (UL) bitrate depends on the resolution and level of compression of the sensor data (or feature data) and can be large.

For object detection and tracking, a video feed is typically uploaded to a server for image recognition and tracking of previously identified objects. The location of the objects, and possibly other metadata, is sent to the HMD. Object tracking places stringent requirements on the latency if the function is offloaded. Offloading only object detection could be less sensitive to latency, as the corresponding information can be sent to the HMD when it becomes available. Then the latency-sensitive

object-tracking functionality on the HMD takes over to ensure that the visual information presented to the user stays aligned with the real world.

In cases where processing is done in the device and/or in a tethered companion device, the communication over the network is limited to application data, with characteristics that are similar to mobile broadband (MBB) traffic. The real-time processing power is mainly limited to what is available in the glasses and the companion device. The device capabilities are much more limited when running without a companion device with local processing only in the HMD. The processing in slim-form-factor AR glasses is expected to only be sufficient for basic SLAM and rendering, while more of the SLAM needs to be performed remotely for immersive and interactive experiences. For the cases with complete offload of all XR application processing functionality, the HMD will always have to run some processing locally, including overlay imaging, localization and latency-hiding processing.

In short, smaller XR device form factors, longer battery life and advanced XR services will require more cloud computing due to the increased need for computation offloading. The impact on network requirements will vary depending on which XR functions and/or their various functionalities need to be offloaded. It makes sense to offload XR functions that enable the highest device power savings or the greatest QoE improvements, as long as the connectivity requirement in terms of bitrates and latencies can be met by the available networks. Offloading will then be progressive as networks evolve and become more capable to meet the various

Key terms

Foveated rendering is a technique that uses eye-tracking to identify where a user is looking and render higher resolution images in those directions, while rendering lower resolution images for peripheral vision. The purpose is to save bandwidth and reduce the rendering complexity.

Object persistence means that a virtual object remains in the same location, even if the user moves out of the environment. A placed object could be stored in a server along with a precise position in a stored map. The map can be tagged with its location in the real world. The storing of detailed maps of scanned environments comes with privacy concerns and is a topic of ongoing research and regulation.

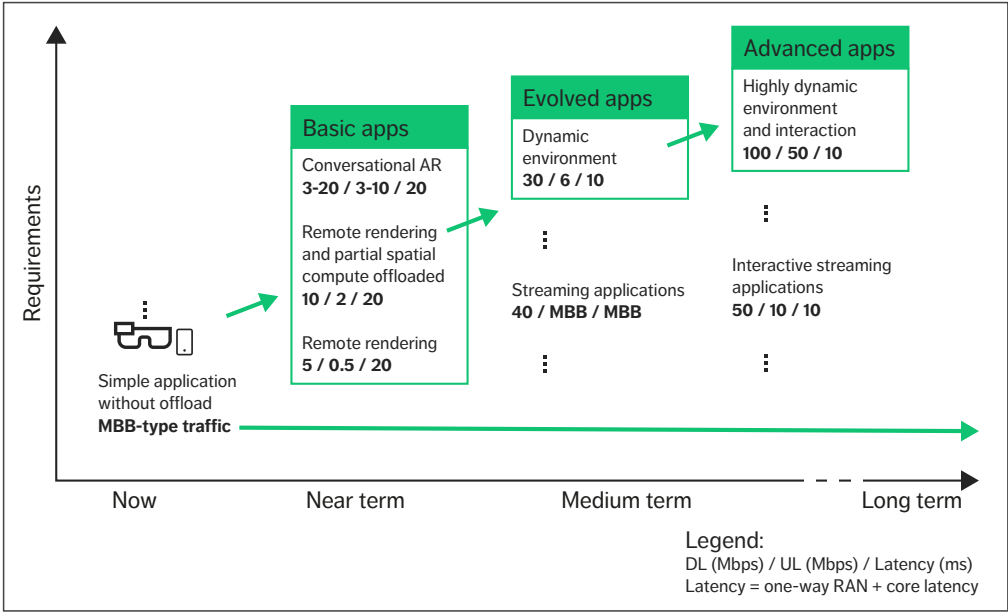


Figure 3 Evolution of requirements over time

and changing requirements of XR services and offloading levels.

Example operating points – XR flavors

A large multi-dimensional space results from the combination of a broad range of more and less advanced XR applications and various XR device capabilities that are associated with different levels of computation offloading. In theory, any point in this space is a possible XR flavor with its own requirements for network support. To simplify the analysis of required network solutions we narrow our focus to a smaller sample. The sample of XR flavors that we present here is based on concrete experiments with various partners [5, 6] and our own research and has been chosen to represent a potential evolution of slim AR glasses. The time dependency of the dimensions is a key consideration.

The XR flavors that we have chosen to analyze should be understood as a limited selection of samples in a wide range. They are examples rather

than forecasts, and the conclusions we draw cannot be directly extrapolated to all XR applications and all cases. The requirements that we have obtained from the different XR flavors are simplified and intended to provide general guidance. In reality, XR requirements will be very elastic in both DL and UL bitrates, as well as in latency, at least to some degree. Thus, it is expected that XR requirements will be defined by a range of values as opposed to a single value.

In Figure 3 we present the XR flavors along a time axis and a strictness-of-requirements axis. For each point in time – now, near term, medium term and long term – there are flavors with different characteristics, resulting in different requirements on the network. Increasingly advanced XR applications will become available over time, which will, together with the device capabilities, place increasingly stringent requirements on the connectivity. Still, at each time instance, a wide range of applications will be used, some of which will be very simple, while others will be more complex. The

THE CONVERSATIONAL AR FLAVOR THAT WE HAVE CONSIDERED IN THE NEAR TERM INVOLVES NO COMPUTATION OFFLOADING

bitrate requirements shown in Figure 3 are aggregated over the flows, whereas the one-way delay requirements are for the most sensitive flow.

Now – simple XR applications without offload

In the simplest XR applications, for which offload provides limited benefit, all processing is done locally on the HMD or on a tethered device. This is similar to how AR applications work on mobile phones today. The requirements are much like those of MBB, and there are no stringent requirements on latency. Example applications include simple navigation services and the ability to extend the touch-screen user interface of smartphones and tablets beyond the bounds of their physical screens with virtual screens controlled by hand gestures. Relevant parts of the global representation of the world are downloaded without real-time requirements. The global map used by these applications is updated on a relatively slow timescale. The time lapse before a newly-created AR object – a new restaurant in a navigation app or a user-added static virtual object in an AR geocaching app, for example – becomes visible for other users depends on how often the global map is updated based on user/actor input and how often the local map is refreshed from the global map.

Near term – basic XR apps

For simple XR applications with remote rendering, we assume that SLAM is run on the HMD. The remote-rendered content would be 2D video content or very simple volumetric content sent in the DL to the HMD. The DL bitrates vary depending on the resolution, frame rate and level of compression. The UL traffic consists of the HMD pose and user input,

which is sent to the remote server to ensure proper rendering. This category of devices would require a relatively low UL rate of 0.5Mbps. A slightly higher DL rate would be needed depending on the size and resolution of the virtual content created in the edge cloud. We chose a sample point of approximately 5Mbps. The one-way latency for such a use case would be estimated at around 20ms, including the radio-access network (RAN), the core and the transport network. In this flavor, exchanging anchor points between users enables the alignment of coordinate systems and local multi-user applications such as board games.

In simple XR applications that offload both rendering and partial spatial computation, localization occurs in the HMD, while spatial mapping, map optimization and object detection are offloaded. This means that 3D objects are rendered on the remote server, and 2D images for the displays are rendered in the HMD. This setup provides the HMD with 3D content and allows for local adjustments in the rendering to account for the user’s head movements as late as possible in the rendering chain. The DL throughput depends on the compression and resolution of the 3D content. We use 10Mbps as an estimate of what could be realistic for low-resolution 3D content.

The UL data is composed of the HMD pose, user input and data representing the detected environment in the form of UL video and sensor data, for example. The UL bitrate depends on the resolution and frame rate of the video and sensor data. A more detailed representation of the detected environment requires better maps and more precise localization. Parts of the spatial mapping and remote rendering have an impact on latency requirements. The offloaded spatial computation functions of this flavor enable an increased understanding of the local environment.

The conversational AR flavor that we have considered in the near term involves no computation offloading. The requirements focus on sending a volumetric data stream generated by one device to receiving HMD(s). At one end, a person is continuously scanned and a volumetric data stream

is generated and sent to the receiving HMD(s) for display. In this scenario, we assume that the rendering from volumetric data to the HMD is performed on the HMD. The UL and DL bitrates depend on the resolution of the point cloud, the frame rate and the level of compression. A reasonable operating range in the near term would be up to 20Mbps in the DL and 10Mbps in the UL. The DL bitrate will depend on the number of participants in the conversation.

Medium term – evolved XR apps

The requirements to support more advanced applications will be more challenging, particularly in dynamic environments where AR objects must adapt to changes in the physical world in real time. One example would be to have a virtual AR character, such as a Pokémon, walk alongside a user on a sidewalk and avoid colliding with anyone or anything around it. Both remote rendering and the offload of spatial mapping will likely be required to achieve this. It will be essential to update the digital representation of the physical environment quickly enough for the application engine to respond with a timely and appropriate rendering of the AR object. The latency requirements in both the UL and DL for this flavor are much tighter than for previous ones, because the consequences of failing to detect and map certain parts of the environment with sufficient quality are greater. For example, the AR experience would be seriously undermined if an object in front of the Pokémon were left undetected and the Pokémon walked through it rather than jumping over it.

To avoid such error events, the representation of the detected environment needs to increase in size (in terms of higher sensor resolution and higher sensor frame rates, for example), which leads to an increase in the UL bitrate compared with previous flavors, based on the assumption that the content sent in the DL is remotely rendered, high-quality 2D video streams.

In the medium term, we also expect to see streaming applications evolve to use XR. For example, content-providing apps such as Netflix, TikTok and Snapchat may provide volumetric video

streams that offer more immersive 3D-viewing experiences such as the Star Wars movie scene where Princess Leia is projected by R2D2 as a hologram. In this flavor, SLAM is handled on the HMD, along with decoding to 2D. The DL requirements would vary depending on the video quality and frame rate. A point cloud stream of 40Mbps could represent two digital objects at a given resolution, for example. The UL bitrate and latency requirements for this flavor are similar to MBB traffic requirements.

Long term – advanced XR apps

In the long term, we expect that interactive streaming applications will have the ability to both stream volumetric video and enable user interactivity – that is, the user’s behavior will affect the content that is streamed from a server. In this flavor, we assume that SLAM will be offloaded. Rendering of the volumetric video stream, based on user interactions, will be performed at a remote server. The DL bitrate will depend on the complexity, resolution and frame rate of the content. The UL will be dominated by the HMD sensor data that is needed to offload SLAM. Low latency will be required both for SLAM offload and for user interactions with the content.

In future advanced XR apps that support highly dynamic environments and interactions, AR objects will be expected to adapt to the dynamic environment, moving as if they were physically present in it. It will also be possible for multiple users to perceive and interact with the AR objects simultaneously. The AR object casts shadows, is occluded by physical objects and is shown correctly

THE AR OBJECT CASTS SHADOWS, IS OCCLUDED BY PHYSICAL OBJECTS AND IS SHOWN CORRECTLY AS A NATURAL PART OF THE PHYSICAL WORLD

as a natural part of the physical world. In this flavor, we assume a dynamic multi-user application with several users in the same environment that interact with each other and the AR objects. The digital world is shared in the cloud.

This XR flavor will require the streaming of high-quality volumetric content in the DL, which could become very large. Alternatively, high resolution and high frame rate remote-rendered 2D video streams could be considered, with or without foveated rendering. To enable correct rendering and spatial understanding, large sets of high-resolution HMD sensor data would be sent in the UL to the cloud. Spatial mapping would be offloaded and correspond to roughly 10-20Mbps in the UL. In addition, high-resolution 3D point-cloud data would be sent in the UL to enable an enhanced experience. This could correspond to 30-40Mbps. Latency requirements would be as strict as those of the dynamic environments flavor in the medium term.

Conclusion

The use of extended reality (XR) technologies is expected to rise significantly in the years ahead, generating new network requirements with varying degrees of stringency depending on device capabilities, how much computation offloading is required, and which specific applications will be used. While these future requirements are heterogeneous and uncertain, they are essential inputs to the process of developing new network capabilities to support future XR use cases. To assist us in this work, we have defined a set of XR flavors to serve as samples in the expected requirements range. Our detailed analysis of these flavors demonstrates that they will require networks that can support bitrates of tens of Mbps together with latencies of 10-20ms to be achieved with high reliability – a task that is significantly more difficult than that of supporting today’s mobile broadband services.

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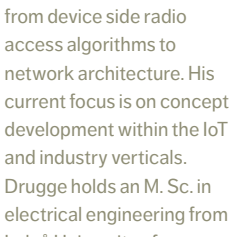
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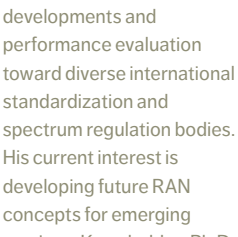
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We would like to thank Håkan Olofsson, Göran Eneroth, Fredric Kronstedt and Yashar Nezami for their valuable contributions to this work.



Network evolution to support extended reality applications

Networks must evolve to handle the traffic generated by extended reality applications. Beyond volume, this new traffic has some important differences compared with mobile broadband traffic. In particular, the need for low latency and high reliability combined with high data rates challenges the capabilities of current mobile networks.

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Enhanced capabilities in terms of both functionality and deployment will be essential for mobile networks to provide a good user experience for extended reality (XR) applications while continuing to deliver high-quality mobile broadband (MBB) and voice services.

As part of the process of defining the new capabilities needed to meet the network requirements of emerging XR applications, Ericsson has defined a set of combinations of XR applications, compute offload levels and device capabilities that we refer to as “XR flavors.” [1] Together with projections on XR uptake, the XR flavors form the

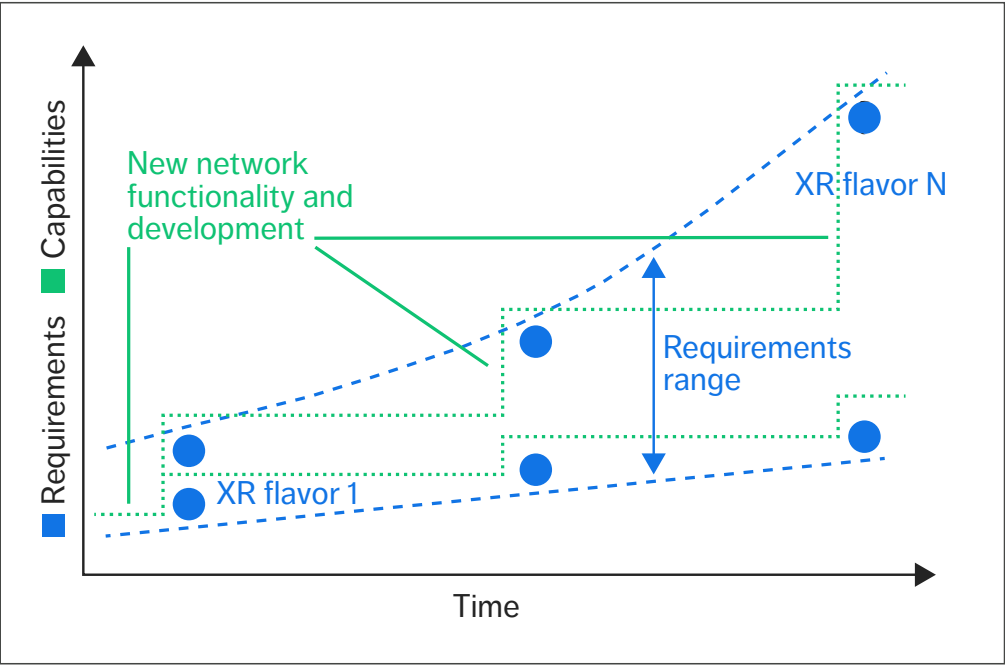


Figure 1 XR flavors in the context of both network requirements and capabilities over time

basis for requirements on network connectivity. The flavors are represented by the dots in Figure 1, which shows them in the context of both increasing XR requirements over time and the new network capabilities that will be necessary to meet those requirements.

The specific traffic characteristics of XR, together with their effect on coverage and capacity, are essential inputs to defining the different kinds of network functionality needed to support XR traffic,

live up to XR requirements and meet the expected performance.

In addition to knowledge about the traffic generated per active XR user and the associated performance requirements, predictions about XR uptake among users and daily usage are also necessary to accurately assess network impacts. These two aspects are difficult to predict, but based on the existing research [2], we assume an uptake of up to 6 percent over the next couple of years. For our

Terms and abbreviations

3GPP – 3rd Generation Partnership Project | **BSR** – Buffer Status Report | **DL** – Downlink | **DRX** – Discontinuous Reception | **FDD** – Frequency Division Duplex | **ISD** – Inter-Site Distance | **MBB** – Mobile Broadband | **Mbps** – Megabits per Second | **ms** – Millisecond | **QoS** – Quality of Service | **RAN** – Radio-Access Network | **TCC** – Time-Critical Communication | **UE** – User Equipment | **UL** – Uplink | **XR** – Extended Reality

●● XR TRAFFIC IS CHARACTERIZED BY MULTIPLE SERVICE FLOWS, ARRIVAL JITTER AND A WIDE VARIATION IN PACKET SIZES ●●

evaluations, we have chosen to use a near-term uptake of 2 percent and a mid-term uptake of 5 percent. With regard to the daily usage, we assume that during the hours of the highest traffic, the usage of the most demanding XR variants (which we use to dimension the network) approaches 10 percent [1]. Active XR users generate traffic according to the bitrates of the XR flavor.

XR traffic characteristics present new challenges

XR traffic is characterized by multiple service flows, arrival jitter and a wide variation in packet sizes [3]. When combined with stringent requirements on bitrates, delay and reliability, these characteristics have a big impact on a network’s ability to support XR services.

XR traffic constitutes several different service flows, including video, audio, pose and control. Different XR applications may use a different number of flows, each with its own traffic characteristics and requirements. For example, pose may have a 10ms latency requirement, while video could have 20ms or 30ms. For good efficiency, the network should adapt to the needs of the individual flows. Today, multiple flows are commonly multiplexed into a single UDP stream, a practice that will need to be revisited to enable better network efficiency.

Arrival jitter is an issue for network support of XR because XR video traffic is typically quasiperiodic, which means that packets arrive at the network with slightly irregular small offsets from the average interval of arrivals. This is caused by delay variations in application encoding and transport networks [4]. The arrival jitter complicates the delivery of the

packets within the bounded latency, as it consumes parts of the delay budget and makes precise allocation of resources in advance more difficult. Arrival jitter also affects several other aspects such as the discontinuous reception (DRX) configuration used to reduce device battery consumption, measurements and scheduling decisions.

The wide variation in packet sizes for XR traffic further complicates the allocation of resources. Depending on the flavor, a substantial part of XR traffic both in the downlink (DL) and the uplink (UL) is video, which is commonly characterized by a mix of packet sizes. Typically, new scenes or larger changes to a scene result in large packets, followed by a series of smaller packets reflecting smaller changes to the scene. In order to handle the largest frames, the network needs to be dimensioned for the peak bitrate. As changes of the scene are not known to the network, frame sizes appear random to the network, which complicates resource allocation.

In short, the combination of the different types of flows, the large packet-size variance even within the same flows, and the arrival jitter, together with the latency and reliability requirement, presents a significant challenge for today’s mobile networks, which are designed for the much more relaxed delay budgets of MBB traffic.

Time-critical communication features for XR traffic

With each new generation, mobile networks have evolved and become more efficient in supporting MBB services. To maintain that efficiency while also ensuring that XR applications work well in 5G networks, it will be essential to separate the XR traffic from best-effort MBB traffic using the 5G Quality of Service (QoS) framework with optimized QoS flows [5].

The 5G QoS framework enables the use of the 5G time-critical communication (TCC) toolbox, which consists of both standardized and non-standardized functionality such as Low Latency, Low Loss, Scalable (L4S) throughput technology [6], UL configured grants, robust link adaptation and tailored scheduling schemes. These tools enable the

optimized treatment of XR throughout the mobile network and specifically in the radio-access network (RAN) to mitigate the different sources of delay. Note that different tools may be applied to different flows.

Even with the help of these tools, however, we can expect lower capacity and worse coverage for some XR flows – and for low-latency, high-reliability services in general – compared with MBB. There are a few reasons for this.

Firstly, fading and interference variations cause the channel quality to vary over time in a way that is difficult to predict. Errors occur when we overestimate the channel quality and select an overly aggressive modulation and coding scheme. In the case of MBB, it is possible to use the more generous delay budget to achieve high efficiency nonetheless. We can wait for instances of good channel quality, use aggressive modulation and coding, and repair errors with multiple retransmissions, if needed.

The situation for XR is different. To support reliable low latency for XR, we need to use more robust link adaptation with an increased safety margin to avoid an overly high frequency of errors on the radio interface and reduce the number of retransmissions. Retransmissions are still essential to efficiently support high reliability, but each retransmission has a delay cost in terms of the round-trip time for sending and responding to a negative acknowledgement. This means that the number of retransmissions required to successfully deliver data needs to be reduced compared with MBB to meet the stricter delay bound. The tendency to underestimate the channel quality directly impacts coverage and capacity. Because of this impact, the level of robustness of the link adaptation should be set according to the requirements and not be overly conservative.

Lower capacity and worse coverage for XR are also caused by protocol aspects. It is not possible to use the whole delay budget to send data, since channel access and retransmissions also need to be accounted for within the delay budget. This means that the rate of data transmission over the air interface must be higher than the data transmission

rate of the application. Due to the arrival jitter and unknown packet size, the data transmission needs to be preceded by a scheduling request and a buffer status report (BSR), in which the user equipment (UE) tells the network that it has something to transmit and how much data it wants to transmit. This takes time, which means that data transmission over the network cannot start the instant the data arrives from the application. The problem can be mitigated by pre-allocating resources by means of configured grants or smart dynamic prescheduling, which the device will use to transmit the BSR. Channel access latency can be reduced, as the UE can skip the scheduling request procedure, leaving more time for data transmissions.

BSR resource allocation accuracy and UE queueing delays are more critical to the capacity of XR service in comparison with MBB. Large and variable packet sizes may cause the UE to report high buffer status indexes that are inaccurate. In these situations the network overshoots the grant size, which often results in resource wastage. The lower the BSR accuracy, the larger the waste, and vice versa. An increased UE-internal queueing delay results when packets with no chance of meeting the latency requirements remain in the buffer for transmission, even if they are not useful for the application, worsening the delays for other packets that could still meet their latency requirements.

Improving standard support for XR

While much of the functionality described in the previous section can be implemented based on current 3rd Generation Partnership Project (3GPP) standards, some would benefit from enhanced standard support. To build an industry consensus, 3GPP has recognized a set of XR requirements and traffic characteristics [4] similar to ours and is using it as the basis for developing improvements. These improvements include XR awareness in RAN, and capacity and power saving enhancements that consider support for high data rates in combination with bounded latency requirements. The XR awareness in RAN is intended to make the RAN aware of both the XR traffic characteristics and the

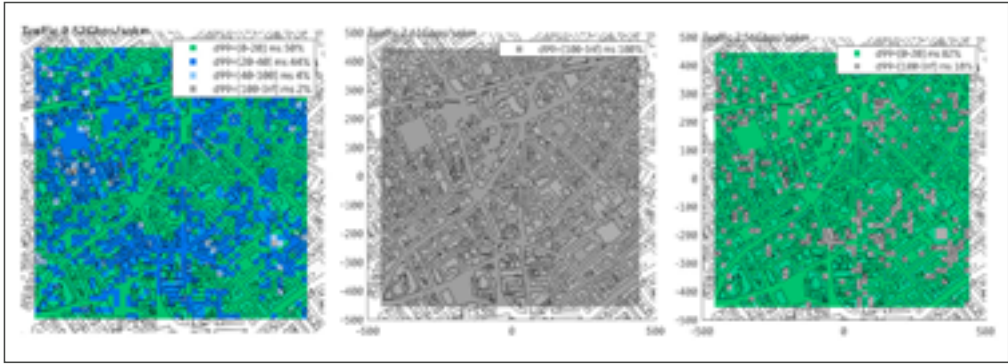


Figure 2 User satisfaction measured as DL packet delay fulfilled with 99 percent reliability

requirements of the XR application, as well as any changes to either of them in terms of the number and types of flows, minimum/maximum/average packet size for each of the flows, flow periodicity, jitter, and the association between IP packets and application packets, for example.

Capacity and power-saving enhancements are also being standardized in 3GPP Release 18. These are grounded in XR awareness in the RAN, which will likely continue to be the baseline for future RAN features specified in 3GPP. The capacity enhancements in 3GPP Release 18 address delay and resource wastage issues by including more precise buffer status reporting with delay information, supporting the dropping of application packets with no chance of meeting delay requirements and studies of dynamic configured grant adaptation.

The power-saving enhancements include adaptations to XR traffic characteristics of DRX, allowing a better matching of monitoring and sleeping periods to the XR traffic characteristics. We expect studying of further enhancements in future releases of the standard.

XR network simulations

To exemplify the ways in which networks will need to evolve to efficiently support XR traffic, we have run several simulations that demonstrate the performance of a range of XR flavors in different

networks with varying characteristics in terms of site density and the type of environment.

Methodology

Our methodology is to first expose a network, as it is, without special functionality or network densification, to XR traffic and measure the XR user experience. In the second step, we introduce TCC functionality to determine the extent to which it can improve the user experience. Then we repeat the procedure for all networks and all the XR flavors [1].

Assumptions

The subscriber density in our simulations is 10,000 subscribers per square kilometer (1,000 for suburban) and 80 percent of them are indoors. Subscribers generate MBB and XR traffic according to the characteristics of the XR flavors and the assumptions on uptake and daily usage. We assume that 70 percent of the traffic is carried by in-building systems such as the Radio Dot System or Wi-Fi. Macro base stations are deployed with different densities in the different networks.

Average inter-site distances (ISDs) range from 150m to 1,000m. 3GPP New Radio is used on low and mid band, with frequency allocations of 2x10MHz frequency division duplex (FDD) at 800MHz, 2x20MHz FDD at 1,800MHz and 2,600MHz, and 1x100MHz time division duplex at 3,500MHz. Massive MIMO (multiple-input,

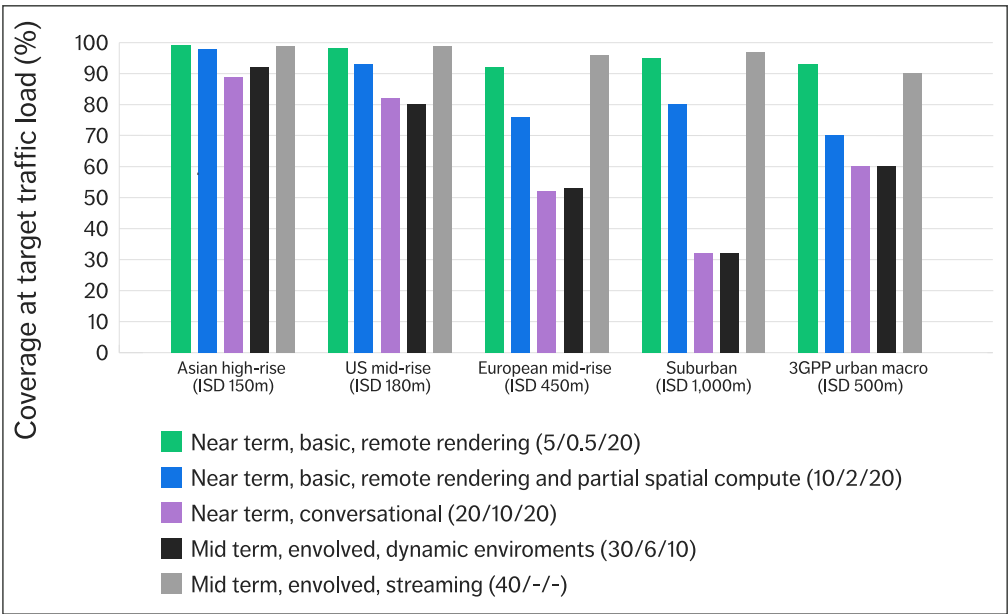


Figure 3 Coverage at target traffic load for five different XR flavors on five different networks

multiple-output) is used on 3,500MHz and regular sector antennas on the other bands. The TCC functionality used is related to service differentiation, admission control, delay-aware scheduling, robust link adaptation and configured grants for BSR. The same functionality is applied to all flows.

Results

Figure 2 illustrates the results for the XR flavor “near-term, basic with remote rendering and partial spatial compute” [1] in a macro network typical for many urban low-rise areas. We present the results as user positions in a three-dimensional map covering one square kilometer, colored by an estimated user experience, where green is good, gray is bad and blue is in between. The user experience is measured as the packet delay that can be supported with 99 percent reliability. A green marker corresponds to a position where 99 percent of the packets are delivered within 20ms, which for this flavor is assumed to lead to an acceptable user experience.

In the case on the left of Figure 2, traffic load is low and no TCC functionality is used. While there are several green markers, the coverage is far from full. In the middle of Figure 2, the traffic load increases to a level that we expect to see in the near future. This causes the map to turn entirely gray, indicating excessive delays, which we attribute to XR traffic queuing behind MBB traffic.

The right side of Figure 2 shows what happens when TCC functionality is added. The map turns predominantly green, with some gray areas that correspond to users who are not served either due to poor coverage or XR overload in those areas. In this scenario, the next step in the network evolution would be the use of capacity and coverage enhancing features on existing sites, or dedicated deployment in those areas.

Figure 3 presents a summary of the results we achieved when we repeated the analysis for five different XR flavors on five different networks that we refer to as Asian high-rise, US mid-rise,

●● DELAY-AWARE
SCHEDULING AND ROBUST
LINK ADAPTATION ARE
EFFECTIVE TECHNIQUES TO
MEET XR REQUIREMENTS ●●

European mid-rise, suburban and 3GPP urban macro. The numbers in brackets represent DL bitrate in megabits per second (Mbps), UL bitrate in Mbps and one-way delay in milliseconds (ms). Coverage is measured as the fraction of users fulfilling the quality requirements in both the DL and UL at the projected traffic load. Three different points in time are assumed, and for each point in time a basic XR flavor and a more advanced XR flavor are covered.

In all five networks, the most basic XR flavor achieves good coverage of between 92 and 99 percent. For other flavors, the coverage varies significantly between the networks. In the Asian high-rise scenario, the network with the highest site density, most of the flavors achieve relatively good coverage, including the more demanding ones. The US mid-rise network is not far behind. Coverage drops for the more sparsely deployed networks, especially for the flavors with higher UL bitrate requirements or lower delay requirements. In these cases, an upgrade of existing sites and a complementary deployment in areas of outage would improve coverage. The impact of densification is roughly indicated by comparing the performance between networks with different site density (represented by the ISD).

Conclusion

The ability to support extended reality (XR) technologies in mobile networks will depend on the availability of new network capabilities that can meet stringent XR requirements on latency, reliability and bitrates. Our research indicates that delay-aware scheduling and robust link adaptation are effective techniques to meet XR requirements in the near

term. Newly standardized features will also assist in meeting the XR requirements.

Our network simulations indicate that many modern networks are already able to support basic XR flavors in terms of coverage and capacity, and that the most densely deployed networks are also likely to be able to support more advanced flavors. For less densely deployed networks, the ability to support advanced flavors will likely require the upgrading of existing sites with enhanced coverage and capacity features, along with targeted densification in the case of the most sparsely deployed networks.

As network densification is complex and time-consuming, XR application developers who want full coverage should be encouraged to develop applications that are designed to adapt to varying network conditions.

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- » Ericsson, Massive MIMO Handbook, available at: <https://www.ericsson.com/en/ran/massive-mimo>
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NETWORKING TRENDS 2023: BUILDING THE PLATFORM FOR NEXT-LEVEL DIGITALIZATION

BY: ERIK EKUDDEN, CTO

One of the most exciting aspects of my career at Ericsson is the opportunity to participate in shaping transformative innovations. Over the course of the last two decades, the tandem evolution of internet technology and mobile networks has played a pivotal role in the digitalization that has utterly transformed both business and society. A high degree of embedded openness has enabled the rapid creation of a wide variety of novel, highly innovative services, and we see that new technologies are being adopted more quickly than ever before. For example, global 5G subscriptions topped the 1 billion milestone at the end of 2022, which is two years faster than the 1 billion milestone for 4G.

The digitalization of industry and society has profoundly changed how most people and organizations consume products and services. As a next step, the mobile industry will transition toward a business platform model that enables the creation of a truly open platform for business and technology innovation. Cloud and artificial intelligence (AI) technologies, programmable networks, application programming interface (API) exposure and as-a-service business models are all instrumental in this process. A network platform providing seamless and

efficient connectivity and communication services will be essential to link consumers, enterprises and society at large.

The network platform will have the ability to handle a large variety of use cases, from best-effort mobile broadband services to guaranteed quality of service (QoS), building a global platform for anyone or anything to consume its services. At the same time, it will minimize the cost and resource usage for these differentiated services, as well helping society and industry to reach net-zero emissions and other sustainability goals.

The network platform is a cornerstone of a future world that is distance-free: a place where the physical and digital worlds coexist seamlessly and where both humans and machines can connect, collaborate and engage with each other regardless of their geographical locations. Universal, resilient and trustworthy connectivity is the keystone of this cyber-physical world.

With all of this in mind, I have chosen to focus on three key trends in this year's article:

1. Technologies that enable a cyber-physical world
2. The need for an open platform for business innovation
3. The high-performing network infrastructure

KEY TERMS

The business innovation platform is open, global and without boundaries, connecting consumers, enterprises and society at large, to enable the creation of new and innovative services that generate exponential business growth. It will be realized by the network platform.

The network platform will aggregate and abstract the network capabilities from the distributed network infrastructure into monetizable, globally available services for developers and enterprises. These services are enhanced by the integration of other value-adding functions originating from the global ecosystem. The network platform embeds interoperability and routing across the actors in the ecosystem to secure service continuity for global scale and reach.

The distributed network infrastructure is evolved with programmability and exposure of network capabilities. The programmable network infrastructure offers service differentiation and deployment flexibility. The network infrastructure capabilities are exposed to the network platform through APIs.





TREND #1

TECHNOLOGIES THAT ENABLE A CYBER-PHYSICAL WORLD

Immersive user experiences such as virtual reality (VR) and augmented reality (AR) have gained a lot of attention in industry and society in recent years. At the same time, technology giants around the world have made significant investments to speed up technology development in this area, which has led to progress in both extended reality (XR) devices and applications.

The evolution of immersive communication that ties user experience to the physical world has been a recurring theme in my CTO Technology Trends articles for several years now. This year, I want to focus on two specific capabilities that are vital for a cyber-physical world. The first, massive twinning, connects and aggregates the physical world of senses, actions and experiences, and uses that information to generate a programmable digital representation. The second, situational awareness, refers to the use of sensing technologies to monitor the surrounding environment, locate objects, and navigate for smooth and safe interaction in a cyber-physical environment.

Massive twinning – real-time digital replication

The creation of a digitalized and programmable physical world without temporal or spatial constraints will require massive twinning – that is, the creation of

accurate, fully synchronized digital twins of everything that exists in the physical world, including humans, physical objects, processes and networked systems. Each digital twin knows its history, can predict its future and has full control of its real-time status. In massive twinning, all digital twins are fully synchronized. Vast amounts of data from the physical world are sent in real time to update the digital twins for use by applications. Actuators and interfaces in the real world carry out functions from the applications for action and experiences.

Massive twinning offers support to minimize resource use and fault prediction. One example of such a use case is a real-time programmable digital twin of an entire smart city that enables precise modeling, monitoring and managing of almost any public or private service, such as utilities, public transportation, public health and environmental monitoring. Another important use case for massive twinning is process optimization in a smart factory to improve productivity, flexibility, quality and sustainability. Massive twinning will also be of great use to communication service providers (CSPs) themselves by making it possible for the CSP to generate a digital representation of the entire network that can be used to simulate network rollout, optimize network operations and initiate predictive maintenance.

Ubiquitous connectivity and integrated sensing are essential components of massive twinning solutions. Massive

twinning can involve hundreds of sensors and actuators per square meter, which means that the network needs to provide advanced service coverage wherever and whenever it is needed to a broad range of devices and applications. This includes the aggregation of data rates up to gigabits per second, which leads to stringent requirements on very low latency and high throughput.

In the 6G time frame, the network will provide connectivity using new and significantly more energy-efficient signaling protocols, and many of the sensors and actuators used in digital twins will be zero-energy devices that do not require any batteries or manual charging. To ensure massive twinning deployments at scale, a fully automated process to handle sensor and actuator devices in bulk will be required. High network reliability and availability, as well as stringent end-to-end data and privacy protection, will also be needed to meet the required levels of trustworthiness.

To fully unlock the potential of zero-energy devices, a holistic approach is needed. For the devices themselves, the energy-harvesting technologies – as well as the packaging and low-power integrated circuit design – will need to be advanced. In collaboration with Massachusetts Institute of Technology (MIT), Ericsson is exploring new materials and ultra-low energy designs to harvest the radio signal itself. ➔

NETWORK DIGITAL TWIN FOR OPTIMIZING 6G NETWORKS

The evolution from today's radio network simulators to network digital twins that utilize 3D gaming and computer-generated imagery technologies enables new features such as high-resolution of city and indoor environments, detailing

surface materials that influence radio frequency propagation as well as the representation of dynamic scene features such as automotive traffic and user mobility. Recent technological advances have improved Ericsson's capability in

several aspects such as visualization, integration, standardization of formats, collaboration and modeling accuracy. Together, these advances enable leaps in our ability to accurately simulate a radio network in a digital twin.

→ **Situational awareness – sensing surroundings and locating objects**

Situational awareness is the ability to understand the environment and its elements in real time and derive meaningful insights in order to make informed decisions and take appropriate actions. This capability is instrumental to building a cyber-physical world and is based on massive twinning data. The digital world is created by processing the sensing and localization information provided by simultaneous location and mapping (SLAM) algorithms. The SLAM information is used for a variety of purposes including AR, 3D mapping, 3D reconstruction and real-time object tracking.

Spatial sensing capabilities will be an integral part of the future network and have been identified as an innovative feature of future 6G systems. The future network will support spatial sensing by reusing network communication signaling, utilizing the same principles as radar technologies to determine the presence, characteristics and trajectory of objects. This solution is referred to as joint communication and sensing (JCAS) and will enable services for evolving use cases such as smart cities, transportation systems and public safety applications.

The network will collect not only JCAS sensor data but also other data from non-JCAS sensors. The network platform has the capability to process the aggregated sensor data to be used for services such as SLAM, navigation and geofencing. The network platform will offer the processed sensor data through network APIs that are easily consumable for the developer ecosystem.

The main advantage of the network platform in terms of future sensing is that most of the network infrastructure is already in place, providing full-area coverage that facilitates a synchronized multi-static sensory mesh. Another useful feature of the network infrastructure is that the radio signals for sensing work in complete darkness and can even sense local weather conditions.

INTEGRATING SENSING INTO NETWORK INFRASTRUCTURES

1 THE EU HORIZON 2020 HEXA-X PROJECT

Together with our partners, Ericsson is engaged in the EU Horizon 2020 Hexa-X project on scenarios and use cases for sensing. The project also includes gap analysis to consider questions including what is needed to support the identified use cases, what can be supported with current 5G technology, and what additional functionalities are required. In the successor project, Horizon Europe Hexa-X-II, the focus will be on sensing architectures for JCAS, physical layer solutions and security aspects.

2 KOMSENS 6G

Ericsson is active in KOMSENS 6G – a research project funded by the German Federal Ministry of Education and Research – where we are working with partners on research that focuses on the integration of sensing in communication networks.

3 NXP SEMICONDUCTORS

Ericsson has teamed up with NXP Semiconductors to investigate potential new use cases for a network where communication and sensing functionalities are fully integrated into the same transmission/reception nodes. Together, we will also investigate implementation aspects of such future systems and evaluate the technical challenges and opportunities for JCAS use cases.



TREND #2

THE NEED FOR AN OPEN PLATFORM FOR BUSINESS INNOVATION

An open platform for business innovation links consumers, enterprises and society at large across sectors and without boundaries. All producers develop and offer their own products, services or content through the platform. The platform will create connections and enable transactions between producers and consumers. Existing platform users, both producers and consumers, will receive a benefit from every new user that enters the platform. As more users join, the advantage for existing users continuously increases. In economics, this is referred to as the “network effect.” An additional advantage for users is that the platform scales without any significant investments.

Inherent to the open platform for business innovation is its capability to support different use case characteristics from the simplest to the most advanced. The services are segmented based on shared requirements and needs to optimize fast deployments and minimize cost. The platform offers flexibility for the differentiation of products, services and content.

In the future, enterprises will consume network connectivity and features as a service that fit their specific needs, similar to the way cloud service works today. Depending on the use case, the service will be handled by a public network or by self-contained and independent logical network slices created in the public network that can extend into purpose-built private networks.

Either way, the result is a virtual network offering that is both secure and dedicated, with a single enterprise management system for enhanced asset control and visibility, including assets connected through the macro network.

The role of the network platform

Today’s network platforms provide communication-as-a-service APIs by, for example, offering SMS, video calling and voice services for developers to integrate into their application offerings. In the not-so-distant future, it will be possible to provide developers with much more advanced and programmable network capabilities tailored to application requirements on latency, reliability and

throughput. These capabilities will be exposed as network APIs tailored for next-generation use cases such as immersive communication and situational awareness. Such future use cases can be further enhanced through the integration of other value-adding functions provided by the global ecosystem.

For the most efficient and scalable implementation, the network platform will aggregate and abstract exposed network APIs from the distributed network infrastructures. Further, the network platform will embed interoperability and routing across the actors in the ecosystem to secure global scale and reach.

The first step in this process will be the transformation of the existing network

infrastructure architecture – including functions such as the radio-access network, the core network, operations support systems and business support systems – by strengthening the network API enablement suite in areas such as resource management and service orchestration. The second step will ensure that the network API enablement suite is equipped with state-of-the-art exposure frameworks and service definition; in other words, a network abstraction layer. As network infrastructure capabilities such as positioning, sensing and embedded compute evolve through 5G Advanced and 6G, the network API enablement suite will build new service segments and policies for service differentiation.

LIVE QoS-ON-DEMAND DEMO AT MOBILE WORLD CONGRESS 2023

Together with three CSPs and a few of our ecosystem partners, we took the opportunity at MWC 2023 to showcase how to make advanced open-network APIs easily available to the developer ecosystem. The demo exposed 5G network APIs to enable QoS-on-demand for interactive HD videoconferencing on mobile devices and lag-free mobile gaming use cases for application developers. The QoS-on-demand demo is based on the telco standardization of 3GPP network APIs and CAMARA communication APIs.



TREND #3

THE HIGH-PERFORMING NETWORK INFRASTRUCTURE

The foundation of modern network infrastructure is an open architecture with open interfaces and open APIs that enable continuous evolution in line with market expectations on business innovation and growth. Limitless connectivity, trustworthy systems and AI-powered cognitive networks will be instrumental technologies in the evolution ahead, from 5G to 5G Advanced, and on to 6G.

Limitless connectivity with greater openness

Open access interfaces, open service interfaces via network APIs, open network

infrastructure layers to cloud, transport and data pipeline interfaces as well as open network applications, such as rApps, are all crucial architectural principles of the network infrastructure.

Service differentiation and deployment flexibility will be enabled with the help of end-to-end managed network slicing that dynamically optimizes the network resources within the stipulated QoS during the network slice life cycle.

Programmable networks will allow for the optimization of the overall network performance as well as the customization of device behavior targeting specific use

cases. Programmable networks build on cloud technologies for faster feature development, faster bug fixing and DevOps-like operations. This will enable CSPs to adopt a software-as-a-service business model for network applications such as billing, management systems and core network functions.

Network-aware applications use knowledge of network capabilities to intelligently place application components where they will run with the greatest efficiency, performance and energy efficiency, taking into consideration factors such as where specific hardware-

acceleration capabilities are available.

An architecture built on distributed network infrastructures also provides limitless connectivity – that is, connectivity that provides the required data rate and latency, and is available wherever needed. Multi-connectivity, multipoint connectivity and distributed MIMO (multiple-input, multiple-output) will be utilized to ensure robustness, performance and consistent quality, which will contribute to increased network reliability, availability and resilience (NRAR). Multipoint connectivity and distributed MIMO provide the ability to use many tightly coordinated network nodes to communicate to each specific device, which greatly enhances performance. To provide truly global coverage, the distributed network infrastructures will include non-terrestrial access components such as satellites and drone networks. As part of the longer-term radio-access evolution, spectrum utilization will extend into the sub-THz frequency bands. These very high frequencies create the possibility of highly accurate sensing for use cases such as JCAS.

Energy-efficient networks will be built by optimizing energy performance end to end. At node level, hardware accelerators and related software components such as operating systems and drivers are essential for low-energy solutions. They will provide scalable traffic processing including features like multiple sleep states and fast activation. Further, smart network and AI-based features for site energy orchestration will be embedded for network energy optimization.

In the 6G time frame, the network infrastructure will be capable of providing more than 100 times higher capacity in comparison with today's network infrastructure, with peak rates over 100Gbps, wide-area consistency latency below 5ms and centimeter-level position

accuracy. The always-available, high-performance network infrastructure will offer truly global coverage in indoor, outdoor and non-terrestrial environments. This is made possible through interoperable service interfaces that interconnect all ecosystem actors on a global scale.

Embedding trustworthiness

The openness of the network infrastructure requires security assurance based on zero-trust principles – never trust and always verify – to secure all communication while constantly monitoring the trustworthiness of the communication. A secure, resilient and privacy-preserving network infrastructure is a cornerstone in supporting all society-, mission- and business-critical use cases for humans and intelligent machines in a verifiable way. This requires seamless integration of AI-based automated security controls and assurance mechanisms to make it possible to respond in real time to both known and unknown threats. This will secure trustworthy network infrastructure that performs tasks reliably, maintains data provenance, confidentiality and privacy, as well as protecting against unauthorized access or manipulation.

In a trustworthy system, secure identities are needed at all layers of the system – infrastructure, connectivity, devices and the edge – as well as in network-slicing functions. This can be enabled by means of root-of-trust mechanisms for identities. To ensure both security and privacy, the network infrastructure will use state-of-the-art protocol stacks including quantum-proof algorithms and confidential computing capabilities to protect data when it is being processed and stored. Finally, the network infrastructure will support security assurance throughout the application communication path, including devices, all network domains and the

application cloud. In the long term, this will include support for bystander and spatial privacy-preserving solutions for immersive user experiences.

Artificial intelligence in programmable networks

Network infrastructure must have the flexibility to handle a large variety of use cases, which will be achieved through greater automation of network management. Programmability and distributed intelligence are the best ways to achieve this. Fully autonomous system operation is based on zero-touch deployment and operation across all stages of the life cycle, as well as on the ability to continuously optimize performance in real time. Network services will be managed by well-defined functional models and by dynamic and fine-grained allocation of heterogenous network resources.

Distribution of AI-based performance optimizations in the network requires a data-driven architecture and data-driven operations. These optimizations are continuously supervised and tuned to deliver explainable optimization results. The network will aggregate the optimization results and place them into a larger context, where they can be combined, evaluated and adjusted to reach a defined intent. A hierarchical control-loop structure is used to manage and optimize the network and end-to-end services. In an autonomous network, humans can take a supervisory role relying on trustworthy AI that is lawful, ethically adherent and technically robust.

Generative AI has reached commercial maturity and has been proven to provide efficiency gains in areas such as research and development. Further, this technology will support network analysis and optimization tasks such as predictive maintenance, fraud detection and network planning.

Conclusion

My vision for the future is a connected and sustainable physical world that is both digitalized and programmable, where humans are supported by digital intelligence and immersive experiences. Already today, we can see the emergence of real-time, programmable and collaborative digital twins for a variety of industrial use cases. As the technology evolves, we will be able to build complete smart cities and global intelligent transport systems. All the data made available by digital twins, massive sensing capabilities and spatial mapping technologies will – together with ubiquitous connectivity and network programmability – form a cornerstone of the digitalization evolution.

The unique capabilities of 5G – including global, trusted and secure access – serve as a foundation for both 5G Advanced and 6G, with the differentiating pillars of limitless connectivity, trustworthy systems and AI-enhanced cognitive networks. 5G will evolve into the network infrastructure of choice because it is the most efficient communication technology able to deliver the required openness, scale and affordability to handle the full range of differentiated services for consumers, developers, enterprises, industries and governments.

Ericsson is a leader in the effort to define 6G, working in close cooperation with ecosystem partners that include industry leaders, academia, standardization bodies and open-source forums. By delivering high-performance networks, enterprise networking and network application programming interfaces for monetization, we will help to create an open business innovation platform that connects consumers, enterprises and society in a cyber-physical world without boundaries. This new global platform is going to open up a myriad of lucrative new business opportunities across the entire ecosystem. We're keen to pursue them – we hope you are too.

ABOUT THE AUTHOR

As Group CTO, Erik Ekdén is responsible for Group Strategy and Technology. His extensive experience of working with technology leadership globally influences the company's strategic decisions and its investments in 5G, 6G, edge computing, artificial intelligence, augmented/virtual reality and the Internet of Things. Ekdén's leadership builds on his decades-long career in technology strategies and industry activities. He joined Ericsson in 1993 and has held various management positions in the company, including Head of Technology Strategy, Chief Technology Officer Americas in Santa Clara (US) and Head of Standardization and Industry. He is also a member of the Royal Swedish Academy of Engineering Sciences and the publisher of Ericsson Technology Review.

ERIK EKUDEN
SENIOR VICE PRESIDENT, CHIEF TECHNOLOGY OFFICER



Further reading

- » **Ericsson, Follow the journey to 6G, available at:**
<https://www.ericsson.com/en/6g>
- » **Ericsson Technology Review Magazine 2023, Spotlight on extended reality, available at:**
<https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/spotlight-on-xr>
- » **Ericsson blog, Five takeaways on the future of 6G XR from the 6G@UT Forum, available at:** <https://www.ericsson.com/en/blog/2023/5/five-research-insights-on-6g-xr>
- » **Ericsson Technology Review, Network digital twins – outlook and opportunities, available at:** <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/network-digital-twins-outlook-and-opportunities>
- » **Ericsson blog, Joint communication and sensing in 6G networks, available at:** <https://www.ericsson.com/en/blog/2021/10/joint-sensing-and-communication-6g>
- » **Ericsson, Zero-energy devices, available at:** <https://www.ericsson.com/en/about-us/new-world-of-possibilities/imagine-possible-perspectives/zero-energy-devices/>
- » **Ericsson, Future technologies for an intelligent society, available at:** <https://www.ericsson.com/en/future-technologies>

Using 3GPP technology for satellite communication

Most satellite communication today is based on proprietary solutions, but that may soon change. Non-terrestrial networks became part of the 3rd Generation Partnership Project standard in Release 17, establishing a strong foundation for direct communication between satellites, smartphones and other types of mass-market user equipment.

SEBASTIAN EULER,
XIAOTIAN FU,
SVEN HELLSTEN,
CHRISTOPHE KEFEDER,
OLOF LIBERG,
EDUARDO MEDEIROS,
ERIK NORDELL,
DAMANJIT SINGH,
PER SYNNERGREN,
ELMAR TROJER,
IOANNIS XIROUCHAKIS

As the rate of adoption of mobile communication technology around the world continues to rise, the goal of using it to provide seamless global coverage to anyone, anywhere, at any time has become increasingly important. This has led to major advances in both terrestrial and non-terrestrial satellite networking technology.

■ Smooth interworking and integration of terrestrial network (TN) and non-terrestrial network (NTN) components is the next logical step on the coverage journey to provide enhanced mobile broadband (eMBB) to consumer smartphones (direct-to-smartphone) and Internet of Things (IoT) use cases.

Integration with satellite networking technologies that can provide coverage in areas that TNs cannot reach would help to deliver resilient services to people and businesses currently unserved in both developed and undeveloped parts of the world, bringing significant social and economic benefits [1].

Beyond the benefits NTN will deliver to smartphones, they will also have the capability to support both industrial and governmental IoT devices for verticals such as automotive, health care, agriculture/forestry, utilities, maritime transport, railways, aeronautic/drone sector, national security and public safety.

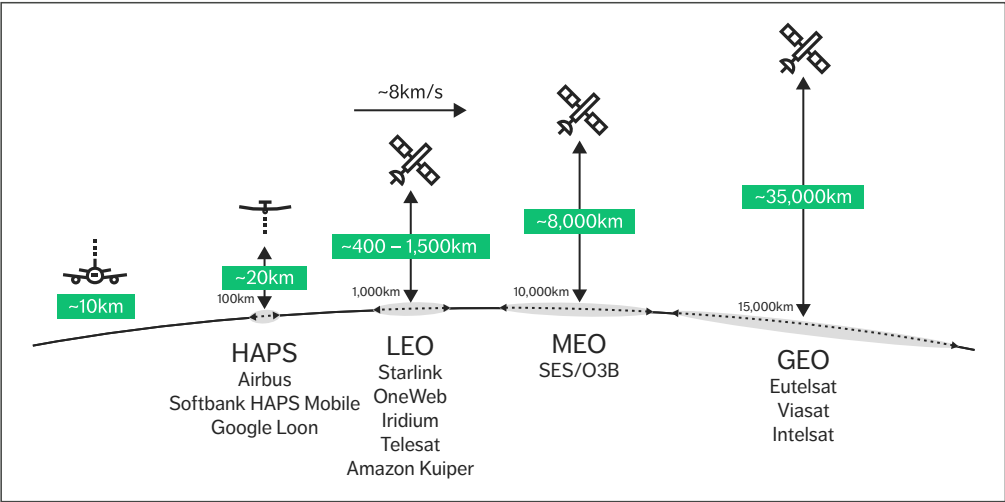


Figure 1 Overview of existing satellite systems

Overview of satellite systems

Different satellite systems have been used for years to provide services such as TV broadcasting, navigation, communications, surveillance, weather forecasting and emergency systems [2]. Figure 1 illustrates the orbits of the three main satellite types – geostationary (GEO), medium-Earth orbit (MEO) and low-Earth orbit (LEO) – in comparison to a

commercial aircraft and high-altitude platform system (HAPS) providing local service coverage.

GEO satellite systems are operated at a high altitude of approximately 36,000km, which introduces long latencies (>500ms) and limited data rates due to the large path loss. GEOs appear stationary to the device and provide a large field of view, which makes them well suited for satellite

Terms and abbreviations

3GPP – 3rd Generation Partnership Project | **5GC** – 5G Core | **CHO** – Conditional Handover | **CSP** – Communication Service Provider | **CT** – Core Network & Terminals | **DL** – Downlink | **FDD** – Frequency Division Duplex | **FSS** – Fixed Satellite Services | **GEO** – Geostationary Orbit | **gNB** – gNodeB | **GNSS** – Global Navigation Satellite System | **HAPS** – High-Altitude Platform System | **HARQ** – Hybrid Automatic Repeat Request | **HRC** – High Reliability Communications | **IoT** – Internet of Things | **LEO** – Low-Earth Orbit | **LTE** – Long-Term Evolution | **LTE-M** – LTE for Machine-Type Communications | **MBB** – Mobile Broadband | **MEO** – Medium-Earth Orbit | **mMTC** – Massive Machine-Type Communications | **MSS** – Mobile Satellite Services | **NB-IoT** – Narrowband IoT | **NG** – Interface between the gNB and the core network | **NMS** – Network Management System | **NR** – New Radio | **NTN** – Non-Terrestrial Network | **ppm** – Parts Per Million | **RF** – Radio Frequency | **RTT** – Round-Trip Time | **SA** – Service & System Aspects | **SI** – Study Item | **SNO** – Satellite Network Operator | **TN** – Terrestrial Network | **UE** – User Equipment | **Uu** – Interface between the gNB and the UE | **UL** – Uplink | **WI** – Work Item

MOBILE SATELLITE SERVICES ARE BEST POSITIONED AS A COMPLEMENT TO TERRESTRIAL MBB SERVICES, CREATING A WIN-WIN-WIN SITUATION

television, business-to-business data services (such as trunking/backhauling and enterprise networking) and governmental services (such as military satellite communication systems).

MEO satellite systems such as Galileo, GPS (Global Positioning System) and GLONASS are mainly used for navigation and are typically deployed at an altitude of approximately 20,000km in a semi-synchronous orbit that is predictable and reliable with an orbital period of 12 hours. There are constellations in MEO that are also used for communications services deployed at a height of about 8,000km. This leads to a latency that is five times lower compared with GEO, providing higher data rates.

LEO satellite systems are used for services such as Starlink, OneWeb, Iridium and Globalstar. These satellites operate at altitudes of approximately 400-2,000km, where a higher speed (about 8km/s, full orbit time of 90-120 minutes) is required to stay in orbit. LEO satellites provide the lowest latency and tens of megabits per second of capacity, making them suitable for MBB and IoT applications. As the footprint is notably smaller compared with MEO and GEO, larger constellations are needed.

Satellite communication use cases and business rationale

The first commercial rocket launches by SpaceX in the mid-2010s coincided with an ongoing paradigm shift in the space industry that soon resulted in a significant drop in the cost of launches as well as an increase in capacity [3]. The “New Space Era” that

has emerged since then has been defined by a dramatic increase in annual private venture capital investments in large LEO constellations focusing on fixed broadband internet services for residential and business users in existing and planned satellite constellations such as Starlink, OneWeb and Amazon Kuiper. The next step in the development of mobile satellite services (MSS) focuses on the ability to communicate with standard smartphones. Three development tracks have emerged: legacy MSS, legacy Long-Term Evolution (LTE) and 5G NTN.

The legacy-MSS track aims to integrate legacy MSS technologies into new smartphones using MSS spectrum. Examples of this approach include Apple iPhone 14 and Globalstar, Huawei Mate 50 and BeiDou, and the addition of Iridium to the Qualcomm Snapdragon Satellite. Meanwhile, the legacy-LTE track is focused on creating a modified network using terrestrial LTE spectrum to reach unmodified LTE phones from LEO satellites. This work is running parallel to the 5G-NTN track, which is based on the 3rd Generation Partnership Project (3GPP) standardized solution specified in Release 17 (Rel-17) that was completed in 2022. Like the legacy-MSS track, the 5G-NTN track uses MSS spectrum.

The assumption is that the legacy-LTE and 5G NTN technologies will be used in cooperation with communication service providers (CSPs), as direct-to-smartphone services cannot compete in terms of price and performance with terrestrial services, where they are available. MSS are therefore best positioned as a complement to terrestrial MBB services, creating a win-win-win situation. This is because, firstly, CSPs run no commercial risk in allowing their subscribers to roam into the satellite network, and doing so enables the CSP to collect roaming fees based on a revenue-sharing model. Secondly, satellite operators gain exposure to a much larger potential market than they are able to reach today with expensive proprietary devices. And finally, the device industry is interested in building devices that are guaranteed to work with satellite systems because they know that this additional functionality will increase the value of the devices.

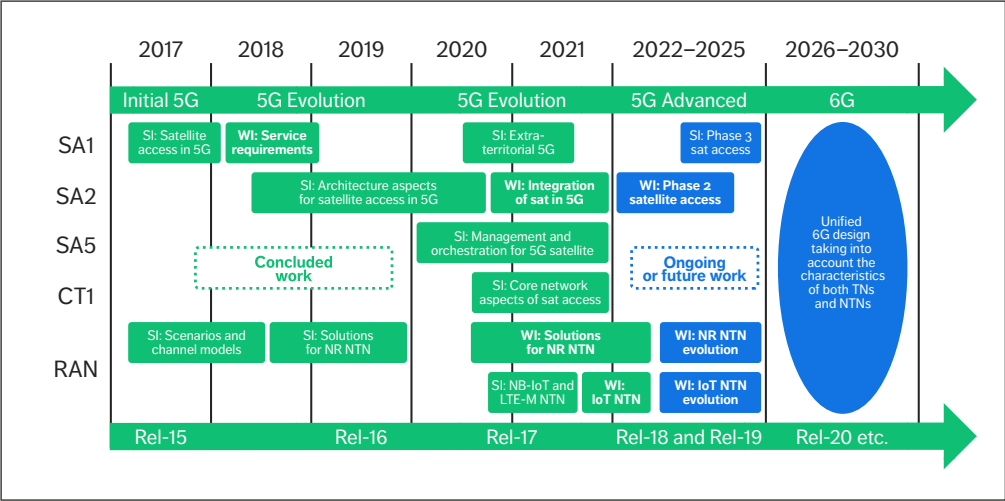


Figure 2 3GPP NTN standardization timeline in the SA, CT and RAN working groups

The 3GPP initiative on non-terrestrial networks

In line with the growing interest in satellite communication in recent years, 3GPP has made efforts to adapt 5G New Radio (NR) as well as narrowband IoT (NB-IoT) and LTE for machine-type communications (LTE-M) to provide satellite-based connectivity. Figure 2 provides an overview of these efforts. Work started with study items (SIs) in Rel-15 and Rel-16, but Rel-17 was the first to include normative work. The focus of the 3GPP NTN efforts so far is on providing communication services to consumers via satellite; other use cases such as backhaul via satellite are out of scope. The work encompasses support for different satellite constellations, in particular LEO above an altitude of 600km and GEO satellites [4].

Figure 3 presents the two different architectures that can be used to realize satellite communication systems based on 3GPP NTN architecture. In general, the satellite radio payload is connected to the core network through a satellite ground station or gateway using what is referred to as the feeder link. The satellite provides communication services to user equipment (UE) via the service link. Although 3GPP Rel-17 specifies the transparent

NTN architecture shown on the left side of Figure 3, the algorithms and enhancements are flexible enough to also support the regenerative architecture shown on the right side.

In the transparent architecture, the base station (gNodeB or gNB) is located on the ground behind the gateway, and the satellite’s main purpose is to act as a repeater. The only processing that can be performed on the satellite is radio frequency (RF) processing such as frequency conversion, amplification and beam management.

In the regenerative architecture, the satellite carries either an entire gNB or parts of it, such as the radio unit which makes it possible to decode and process packets on the satellite. The feeder link in this case is akin to terrestrial fronthaul/backhaul and it is not necessarily implemented using NR. The regenerative architecture provides more flexibility, better performance and global coverage due to the ability to support inter-satellite links.

Rel-17 NR non-terrestrial networks

Modern satellites typically divide their service areas into several hundred sub-areas, which they serve with individual beams (“spot beams”). In general,

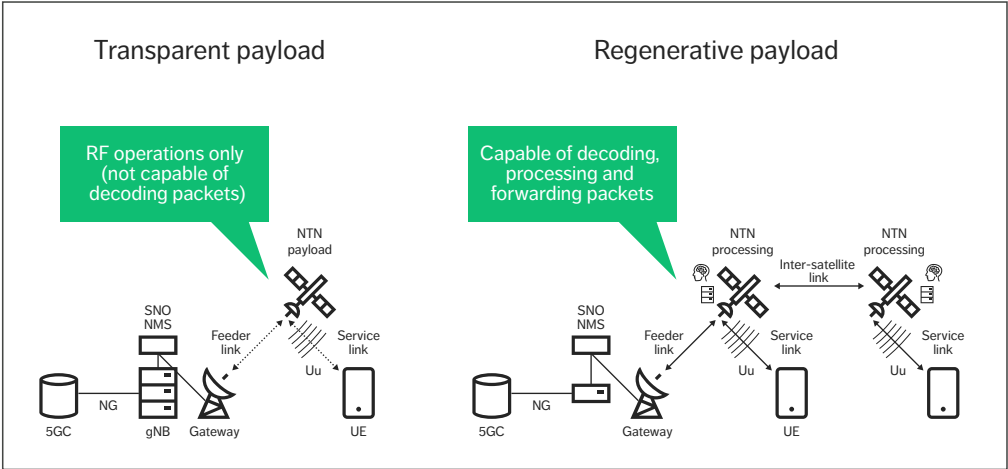


Figure 3 Two different NTN architectures

each of these areas corresponds to one cell, and can have a diameter of tens or even hundreds of kilometers.

While GEO satellites are (almost) stationary with respect to a point on the Earth’s surface, LEO satellites move at approximately 8km/s (~30,000km/h) in their orbits. If the beams are fixed with respect to the satellite, the beams will sweep the surface of the Earth, leading to frequent mobility events, such as handover between cells, even for stationary UEs (typically every few seconds).

Alternatively, a beam steering mechanism can be implemented on the satellite to steer the beams toward a fixed area on the Earth for as long as possible. This concept, known as “Earth-fixed beams,” allows a device to remain in the same beam and cell for several minutes. While both alternatives are supported in Rel-17, a particular benefit of the Earth-fixed beam concept is that it avoids frequent handover between cells.

The fundamental challenge for any satellite communication system is how to overcome the large round-trip delays and frequency shifts due to the movement of the satellite relative to Earth, also known as Doppler shifts. For GEO satellites, the round-trip delay can be longer than 500ms, and even

for LEO satellites it can amount to tens of milliseconds. The differential delay within a cell is also large, extending to as much as 10ms depending on cell size. The fast movement of LEO satellites creates large Doppler shifts of up to 25ppm (50kHz at 2GHz carrier frequency).

The 3GPP solution to this challenge is to require the UEs to compensate delay and service link Doppler shift before accessing the network. To this end, the satellite broadcasts its ephemeris corresponding to its position and velocity. The UE is required to be equipped with a Global Navigation Satellite System (GNSS) module, which it uses to determine its own position before accessing the network.

From its own position and the satellite ephemeris, the UE calculates the distance to and relative velocity of the satellite, and it determines the required pre-compensation values and applies a large frequency shift and timing advance. This enables the gNB to operate at its nominal frequency and with uplink (UL) and downlink (DL) timing aligned, as in a TN.

The long propagation delays necessitate further changes. Scheduling timing relationships, which are designed to cater for round-trip times (RTTs) below

1ms in a TN, have been redesigned to cope with the longer delays as well.

Hybrid automatic repeat request (HARQ) operation, which guarantees reliable data transmission, is also affected. HARQ is a “stop-and-wait” protocol, meaning that a HARQ process ID can be reused only after the corresponding feedback has been received. In legacy NR, there are 16 HARQ process IDs, which in NTN would lead to a situation where no new data can be transmitted simply because there are no free HARQ process IDs available.

To avoid this effect, known as “HARQ stalling,” the number of HARQ processes has been increased to 32. For GEO scenarios with their extremely long round-trip delays of several hundreds of milliseconds, an unfeasible number of HARQ process IDs would be needed. The option to disable HARQ feedback (per HARQ process ID) was therefore also added. In this case, retransmissions are handled by the slower feedback loop that is supported by the Radio Link Control layer.

Mobility is another area in which NTN differ significantly from TNs, most obviously in LEO constellations, where even stationary UEs will experience frequent handovers because of the orbital movement of the satellites. In TNs, UEs experience a clear difference in measured signal strength depending on the distance between the UE and the base station, whereas in NTN all UEs have approximately the same distance to the satellite, with only a small difference in signal strength between cell center and cell edge.

This difference is utilized in legacy mobility procedures such as cell selection, which is based on received DL signal strength. In NTN, the primary solution for connected mode mobility is expected to be conditional handover (CHO). To support NTN, CHO has been upgraded to include a time-based trigger condition and a UE-location-based trigger condition. The former allows the UE to execute the handover during a certain time period, while the latter takes the device location relative to the target and source cells into consideration for the handover decision.

MOBILITY IS ANOTHER AREA IN WHICH NON-TERRESTRIAL NETWORKS DIFFER SIGNIFICANTLY FROM TERRESTRIAL NETWORKS

Other enhancements include support for DL transmission polarization signaling, extension or offset start of various timers, enhancements for cell selection/reselection, reporting of the applied timing advance during random access, and UE location reporting to facilitate procedures like selection of a core network in the correct country and lawful intercept.

Rel-17 IoT non-terrestrial networks

Rel-17 also includes adaptations to NB-IoT and LTE-M that will enable them to support NTN. This 3GPP track is known as IoT NTN. The work item (WI) was started very late in Rel-17 with minimal scope, focusing on essential functionalities. The general approach in IoT NTN is to follow the NR NTN work as closely as possible and adapt its solutions. For example, the basic solution for pre-compensation of delay and Doppler shift is the same, requiring IoT NTN UEs to have GNSS support. The NR NTN enhancements to scheduling timing relationships have also been adopted for IoT NTN. No mobility enhancements (such as CHO) have been considered, however.

Discontinuous coverage is a topic that is specific to IoT NTN. In contrast to NR NTN, the UEs in many IoT NTN use cases may not need continuous coverage. For example, it may be sufficient if they can transmit their data once every few hours. These types of use cases could make it feasible for some satellite operators to deploy sparse constellations with fewer satellites. To support such operation, information – such as the satellite ephemeris of neighboring cells along with coverage info of the cells – will be signaled to enable the UEs to predict the times when they will have coverage.

THE ITU-R VISION FOR mMTC INCLUDES SUPPORT FOR UP TO 500 DEVICES PER SQUARE KILOMETER

Rel-18

Both the NR NTN and the IoT NTN work continues in Rel-18. For NR NTN, the new objectives focus on coverage enhancements, further improvements to the mobility procedures and methods for the network to independently verify the reported UE location. For IoT NTN, the scope includes a method to disable HARQ feedback similar to NR NTN, mobility enhancements such as CHO for LTE-M and further enhancements for discontinuous coverage.

Spectrum for 3GPP non-terrestrial networks

Spectrum for satellite communications is divided into spectrum for providing MSS and fixed satellite services (FSS). The S- and L-bands are examples that belong to the MSS domain, while the Ka- and Ku-bands provide FSS.

Rel-17 specified support for the L- and S-bands as band n255 (1,626.5MHz-1,660.5MHz and 1,525MHz-1,559MHz for UL and DL, respectively) and n256 (1,980MHz-2,010MHz and 2,170MHz-2,200MHz for UL and DL, respectively). Each of these frequency division duplex (FDD) bands offers approximately 30MHz of spectrum in each link direction.

In Rel-18, another MSS FDD band will be added with the UL in L-band (1,610-1,626.5MHz) and the DL in S-band (2,483.5-2,500 MHz). This addition makes about 80MHz of DL and 80MHz of UL spectrum that is suitable for providing operation from a satellite direct to a handheld device available to 5G NR NTNs [6].

In Rel-18, 3GPP will also specify at least three example bands (n510-n512) in the Ka frequency range 17.7-20.2GHz (DL) and 27.5-30GHz (UL). While the L/S-band targets handheld devices, in the Ka-band, devices with higher-gain antennas are

required, such as very small aperture terminals, which are typically mounted on buildings or vehicles.

5G non-terrestrial network system performance

Similar to terrestrial 5G, NTN aims to provide services beyond MBB. The ITU-R (International Telecommunication Union – Radiocommunication Sector) outlines performance requirements intended to facilitate ubiquitous and resilient coverage for MBB, massive machine-type communications (mMTC) and high reliability communications (HRC) [7]. The report elaborates on the key performance requirements for each use case in the context of a LEO 600km constellation operating over a 30MHz carrier.

Notable requirements are peak data rates of 70Mbps (DL) and 2Mbps (UL), corresponding to spectral efficiencies of 3bps/Hz (DL) and 1.5bps/Hz (UL). This can be translated to DL and UL area traffic capacity of 8kbps/km2 and 1.5kbps/km2 for this particular constellation. The ITU-R vision for mMTC includes support for up to 500 devices per square kilometer, while the HRC use case is associated with a reliability of 99.9 percent.

Providing direct-to-handheld connectivity from a satellite constellation presents a considerable challenge due to the propagation loss between the satellite and the handheld device on the ground. The available link budget depends on many factors such as orbit height, system architecture, antenna design, area to be served by the constellation and the elevation angle between the satellite and the UE.

The link budget in turn determines many performance figures, such as the achievable throughput. As an example, a user can expect the highest throughput when the satellite is directly overhead, at an elevation angle of 90 degrees. The path loss increases when the satellite is at lower elevation angles, and at elevation angles around 30 degrees, the achievable throughput might be reduced by a factor of 2.

A second effect of the distance between the terminal and the satellite radio is the increase in latency. The overall latency that the user experiences

will vary depending on the satellite’s position in relation to the user and the ground station. Taking a terrestrial system as reference with a maximum RTT of approximately 1ms, the round-trip latency for a LEO constellation at an altitude of 600km will be between 8ms and approximately 20ms when using the transparent architecture, assuming a minimum elevation angle of about 30 degrees.

Key benefits of a 3GPP-based non-terrestrial network solution

Regardless of the architecture, the main benefit of a 3GPP-compliant NTN solution will be immediate compatibility with mass-market smartphones. In comparison with the bulky and expensive terminals used in non-3GPP-based legacy-MSS systems, a 3GPP solution will bring global data-and-voice connectivity to regular-sized smartphones. Terrestrial operators can boost their geographical coverage and close the gap of connectivity in sparsely populated areas, including rural settings, while reaching new use cases such as maritime coverage.

A system based on 3GPP NTN accounts for the Doppler shifts and delays that are inherent to satellite systems without relying on proprietary workarounds. It is a future-proof solution with an evolution that follows the 3GPP releases and is both backward-compatible with 4G LTE IoT NTNs and forward-compatible with 6G NTNs. It is also very flexible, with the ability to provide connectivity for LEO, MEO and GEO during the full life cycle of a constellation.

A 3GPP-compliant NTN solution makes it possible to deliver a single network that comprises both terrestrial and non-terrestrial components, incorporating the world’s largest ICT ecosystem. This evolution will enable satellite operators to provide affordable satellite communication and inherently better performance due to specified enhancements including the improved HARQ mechanism. And as the services that a 3GPP-compliant NTN solution will deliver will be deployed through NTN-specific spectrum, there is minimal risk of interference.

Conclusion

Satellite connectivity based on open 3rd Generation Partnership Project (3GPP) specifications offers the best opportunity to create a large non-terrestrial network (NTN) ecosystem, enabling connectivity between terrestrial systems and satellite systems on the same mobile platform. As satellite systems will not have the same capacity as terrestrial systems, they should be viewed as complementary rather than competing systems. We expect to see more cooperation between satellite operators and terrestrial communication service providers (CSPs) in the years ahead to achieve mutual benefits in this area.

One of the main challenges that must be overcome in the work to create NTNs is the issue of the interference that can arise if the same spectrum is used for both terrestrial and satellite systems. In many regions, there are already regulatory requirements that could prevent the use of CSPs’ terrestrial spectrum for satellite operations. To overcome the challenge of potential interference, Ericsson’s position is that it is preferable for satellite-based services to use specific satellite spectrum.

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Autonomous networks with multi-layer, intent-based operation

A truly autonomous network has the ability to change requirements dynamically without human involvement. Creating a network that can do this requires the use of intents between multiple operational domains of the network architecture, including the business, service and resource layers.

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As consumer expectations continue to rise and networks grow ever more complex, communication service providers (CSPs) around the world are becoming increasingly aware of the need for intent-based autonomous networks that significantly reduce the need for human intervention.

■ Intent is the expression of declarative requirements that stem from the specific business needs of a CSP and its customers. When operations are based on intents as opposed to tasks, the

THE RECEIVING SYSTEM HAS A MUCH HIGHER DEGREE OF FREEDOM IN CHOOSING WHICH SOLUTION STRATEGIES AND ACTIONS TO APPLY

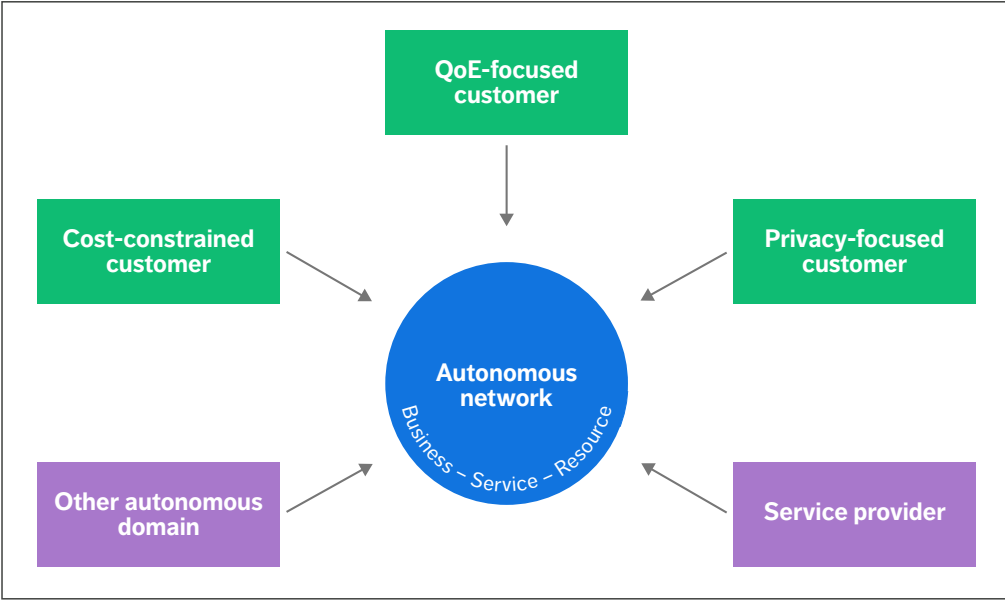


Figure 1 Intents are derived from the business needs of multiple parties within an autonomous network

receiving system has a much higher degree of freedom in choosing which solution strategies and actions to apply. Requirement-centric interaction is essential to achieve a clear separation of concerns, which ultimately enables higher levels of autonomy.

Managing requirements with intents

Most CSPs today have a broad customer base comprised of organizations whose priorities, expectations and network requirements may vary widely. Figure 1 illustrates that one customer may be particularly constrained by cost, for example, while another prioritizes quality of experience (QoE)

above all else. For yet another customer, privacy may be of the utmost importance. The CSP serving these customers will also have its own distinct expectations. As the owner of the autonomous network, the CSP expects it to generate revenue while also being secure, energy efficient and compliant to regulation. The purpose of intents in this scenario is to serve as information objects within the operational processes that carry knowledge about the individual requirements of all parties.

The intent receiver is an automated software system. An autonomous network consists of multiple automated systems arranged in an architecture with

Terms and abbreviations	
API – Application Programming Interface	BSS – Business Support Systems
CSP – Communication Service Provider	DT – Digital Twin
E2E – End-to-End	IMF – Intent Management Function
KPI – Key Performance Indicator	ML – Machine Learning
NF – Network Function	OSS – Operations Support Systems
QoE – Quality of Experience	RAN – Radio-Access Network
SKPI – Service KPI	SLA – Service Level Agreement
SMO – Service Management and Orchestration	

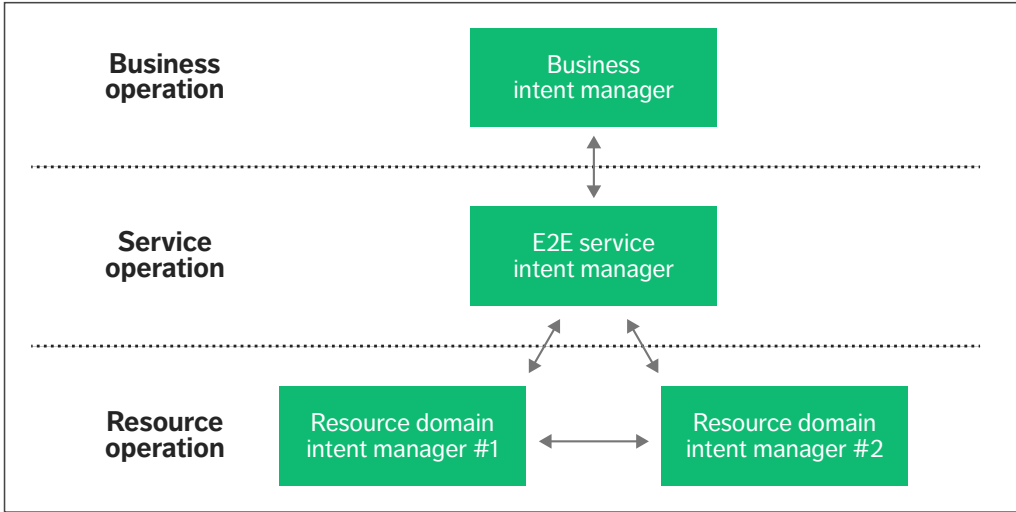


Figure 2 Ericsson's multi-layer, intent-based operation concept

business, service and resource layers, as shown in Figure 2. Each layer contains various autonomous domains [1]. For example, in the resource layer, a radio-access network (RAN) management domain controls RAN resources and services next to other domains, such as transport and cloud management. In this architectural structure, an intent can be used between any operational layers and individual domains. Intent is therefore not only used for expressing the requirements of users and businesses directly, but also for distributing derived individual requirements between operational domains.

A system domain that is able to assure and optimize its operation autonomously within its defined scope of responsibility is referred to as an “autonomous domain.” Within an autonomous domain, an intent manager – also known as an Intent Management Function (IMF) [2] – is the entity that receives and sends intent. It controls the assurance loop for domain-specific requirements within the domain and coordinates with other domain-specific functions regarding which actions to take for provisioning, optimizing and healing. One example of a possible action that an autonomous domain can take in an intent-driven network is sending further intents to other autonomous domains.

The originator of the intent typically has its own requirements to fulfill. To be able to do this successfully, it defines instrumental requirements to which it needs other domains to comply. It uses intent to express these requirements and an intent application programming interface (API) [3] to communicate them and manage their life cycle. The intent receiver needs to assure that its domain complies with the requirements and reports its success back to the originator.

A key characteristic of intent-driven autonomous networks is therefore a hierarchy of requirements expressed through intent and handled through a hierarchy of intent managers and their autonomous domains. Broader requirements with network-wide, end-to-end (E2E) scope are handled in higher layers. They are typically translated into requirements that are specific to autonomous domains. An autonomous domain is responsible for subsets of resources, considering geographical distinction or branches in the network topology, for example. The central task of each individual domain is to ensure it meets its specific intent requirements and thus contributes to the success of the entire autonomous network in meeting CSP and customer expectations.

Achieving a high degree of autonomy

Three factors are essential to reaching a high degree of system autonomy in intent-based operations: (1) assurance, (2) utility-based decisions and (3) control of the effect and risk of actions.

Assurance

Intent managers assure intent-specified requirements by continuously monitoring the system state and ensuring that all requirements are met, even in cases where there are multiple, potentially conflicting, intents. If an intent manager discovers that a requirement is not being met, it will autonomously take countermeasures and inform the intent owner if it cannot find a solution. If a new intent adds requirements, the intent assurance loop will immediately consider them. In this respect, starting a provisioning process is a countermeasure taken by the intent manager’s assurance loop, if it determines that insufficient resources make it impossible to meet intent requirements. If an intent is removed, the intent manager would detect a use of resources that is not justified by requirements. In this case, releasing resources would be a sensible optimization action.

Utility-based decisions

Utility is information about the relative value of an observed state or reached key performance indicator (KPI) value. For example, a typical intent requirement in the service and resource layer may state that a service latency experienced by a user must always be below 100ms. This is a binary requirement that is either met by the system or not. Utility is a gradual expression of the value and importance of any potentially reached latency. If the observed latency is 110ms, the requirement as such is not met, but utility information would quantify if it constituted a slight inconvenience represented by a slightly lower utility value than reaching the 100ms. On the other hand, if the observed latency value represents an important service disruption, the associated utility value would be significantly lower.

Utility is expressed by utility functions and included in the intent expression, which enables the

Key concepts

Intent introduces a structure by which requirements are expressed and managed in a standardized and unified fashion across multiple systems. The requirement-centric approach constitutes a clear separation of concerns between various system layers and domains.

Utility is a measure of business value and relevance of reaching a certain compliance level to a requirement. It will become a key concept for self-adaptive autonomy. The automated system can use utility to make decisions based on a general understanding of value. It has the potential to eliminate the need for use-case-specific logic or human assistance through updating policies. Decisions made through utility maximization ensure that whatever an autonomous domain is doing, it will always be optimized for business value regardless of any new situations or conflicts the intent-based system is facing.

Digital twins are software components that behave like the resources they represent. In the context of intent management, they allow the safe exploration of alternative solutions before they are applied. The combination of exploration with DTs and utility-based evaluation will be a key enabler of autonomy.

originator of intent to not only state requirements, but also express the importance of and preference for an outcome.

The concept can be applied to any requirement on any layer. A utility function for revenue requirements has a direct relationship to business value. A utility function for latency in the resource layer would be shaped with respect to the latency’s indirect contribution to revenue. It can furthermore consider customer value and priority through elevated utility for some requirements compared with others.

This makes utility information the base for optimization and prioritization decisions while facing conflicts. For example, optimizing the system for requirements related to energy saving and user experience often requires opposing actions. Utility

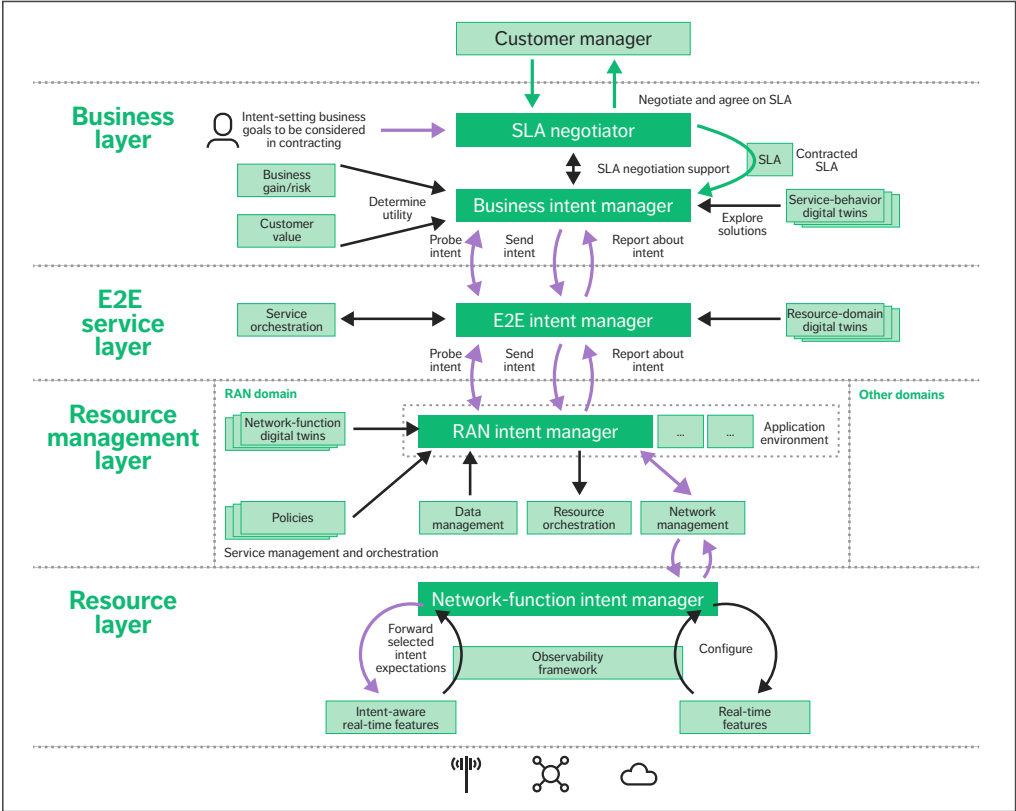


Figure 3 Multi-layer, intent-based operation – detailed view

functions allow for finding the optimal compromise by going for system configurations that maximize accumulated utility across both requirements. The autonomous network therefore reaches the most valuable combined user experience and energy-saving levels possible.

Control of the effect and risk of actions
An intent-based system needs to ensure that its actions not only have the desired outcome, but also have low risk for negative or collateral effects. This can be controlled by employing models that predict possible outcomes and their likelihood. An essential tool would be a digital twin (DT) allowing the

provisional execution of actions without impacting the physical and operational resources. Utility scoring applied to the current system state and possible outcomes of investigated actions would identify and eliminate actions that do not improve the state or bear an unreasonable risk of degradation.
If an intent manager implements intent assurance loops with these capabilities, the result is a highly capable autonomous domain. It would have tight and fully automated control over the domain's compliance to intent requirements, including automated adaptation of intent changes. Through utility-based decisions in combination with

prediction of action effects and risks, it can self-adapt to new situations and always find sensible solutions even for use cases and states that were not explicitly considered at design time. This therefore further reduces the need for human intervention and escalations.

Ericsson's comprehensive, multi-layer approach
The standardization of intent for various autonomous network domains is currently underway in several forums including TM Forum, the 3GPP (3rd Generation Partnership Project) and the O-RAN Alliance. While standards are maturing, Ericsson is exploring the use of intent across our portfolio. *Figure 3* illustrates our approach to intent-based operation use cases in business, service and resource domains, which is based on the cognitive loop presented in a recent Ericsson Technology Review article [4].

Intent in the business layer
The central concern of a business support system (BSS) is to help the CSP capitalize on the investments it makes in its network infrastructure, partner ecosystem, business models and multichannel engagements, while simultaneously maximizing its profitability and delivering a good experience for consumers and enterprises. The decisions taken to realize the desired business outcomes can be seen as reasoning under uncertainty. Intent-based management is well suited to operate in this environment.
In the following examples, we explicitly state the use of the intent-based interfaces [3] between the BSS and the operations support system (OSS). It should be noted, however, that the intent interfaces can also be used between the BSS components.

Part 1: Contract negotiation, operational readiness, and support
In a B2B scenario, every contract between the CSP and a specific customer is a unique instance. Each customer places its requirements on the equipment to obtain a desired business outcome. The customer

WHILE STANDARDS ARE MATURING, ERICSSON IS EXPLORING THE USE OF INTENT ACROSS OUR PORTFOLIO

also has a solid business understanding of how to prioritize the different parts of its business if a resource shortage appears. The choice can be between different equipment using the same service or between different services and the equipment they use. This type of business is driven by Service Level Agreements (SLAs). The SLAs and their related KPIs are business requirements that must be fulfilled with the help of resources in the network.
During the contract negotiation, the customer defines its requirements for the service in the form of service KPIs (SKPIs). With the help of the BSS knowledge and network information, the effects of these SKPIs can be estimated through simulations. Examples of the effects are:

- » How different SKPIs affect each other
- » How SKPIs affect resources
- » How constraints on resources affect the SKPIs
- » How states affect SKPIs.

This additional information can be used to estimate the costs of the contract, the contract implications, and the probability of an impact on the existing resources. The estimates may lead to a variety of possible outcomes including a price modification of the service, a decrease in the amount of user equipment, network buildout or constraints on the stated SLA.
Business metrics are used to calculate the utility of the service(s) covered by the contract, which is defined in terms of the business importance of the service. Examples of metrics include the cost of SLA violations, economical margins, the business state of the customer and the importance of a long-term relationship. The business importance of the service

●● TOGETHER WITH A SERVICE ORCHESTRATOR, AN E2E INTENT MANAGER IS ONE OF THE KEY COMPONENTS OF AN INTENT-AWARE SERVICE LAYER ●●

can be updated at any time, which triggers a reevaluation of the network capacity and business risks.

The information in the contract proposal, together with the calculated business importance, is used to estimate and simulate the effects of the contract on other customers' services during normal load and during resource shortage. The estimated effects and the probability of their occurrence are used to analyze the business risks of accepting the contract. The estimations and simulations are supported by the BSS knowledge.

After the simulations, requests are sent to the OSS and the SLA monitoring function to get a more accurate estimate of the abilities to honor the SLAs and SKPIs. When detected, issues in the contract are corrected and agreed upon, and requests are sent to the OSS and the SLA monitoring function. Both requests are sent through the IMF interface as intents.

Part 2: Contracts in operation

During the life cycle of an agreed-upon contract, its SLAs must be monitored, and potential SLA violations detected and mitigated. An SLA-monitoring function within the intent manager's control loop predicts and detects violations. Further functions in the loop propose mitigation strategies and the probability of their success, which are analyzed from a business-risk perspective before deciding on actions.

While it is not possible to alter an SLA without engaging in contract negotiations with the affected customer, the intent manager can resolve performance issues without the need for contract negotiations by proposing solutions that are adjusted

for business importance. These adjustments are based on the result of estimations and simulations supported by the BSS knowledge. Adjustments to business importance are expressed through utility information within intent and sent to the service layer through the intent interface.

Intent in the end-to-end service layer

The primary purpose of the service-layer intent manager is the realization and assurance of E2E service in accordance with service requirements received from the business layer. Compliance to intent received from the business layer is the terminal goal of the service layer. Central to achieving this is the translation of service intent into instrumental goals constituting resource-layer requirements for individual resource domains. The idea is that the business intent is met as long as each individual resource domain remains compliant to its assigned intent.

Together with a service orchestrator, an E2E intent manager is one of the key components of an intent-aware service layer. This intent manager analyzes the requirements regarding services to be delivered to customers and decides which services exposed by the network would be suitable. Digital twins of resource domains can help in the evaluation of potential solutions [6]. The intent manager orders the selected services from the orchestrator resulting in various actions toward resource domains. The E2E intent manager also monitors the continued fulfillment of the customer requirements.

With intent-aware resource-layer domains, the action that the service layer takes can be the sending of one or several coordinated intents. This can be in addition to other actions using non-intent-based interfaces for configuration, provisioning and management. While this article focuses on the RAN domain, coordinated actions would also be taken in other domains such as core network or transport. If the respective resource domain is intent aware, such action can involve the sending or resource-specific requirements through intent.

Intent sent into the RAN domain typically involves quality-of-service-related requirements

such as experienced service latency and available throughput. For example, the latency requirements sent to the RAN domain constitute the latency budget that the RAN has to comply with as part of the overall E2E service latency. This breakdown from E2E requirements into partial contributions is a central task of service-layer intent management. The intent can be further contextualized, for example, regarding timeframe of validity, geographical region or user groups.

Intent in the RAN domain: RAN management

The central task of RAN management is to assure that the performance of the RAN infrastructure meets the requirements that the E2E service layer has allocated to it. This article considers a RAN resource domain with two layers of intent. The intent coming from the E2E service layer is handled within the RAN management layer. We have chosen to define the RAN resource management layer according to the system architecture of Service Management and Orchestration (SMO) as defined by the O-RAN Alliance [5]. This architecture is characterized by a set of common platform functions including data management, analytics, orchestration and network management. Furthermore, it provides execution and management environments for applications, policies and machine learning (ML) models.

The applications in the automation platform are software components that implement use-case-specific logic and network control, interacting with common platform functions through standardized interfaces. Based on these principles, we have created a RAN intent manager as an application within the automation platform.

As shown in Figure 3, the RAN intent manager receives intent from the E2E service layer. The intent manager verifies compliance based on observations acquired through data management and analytics within the framework of the automation platform. If it identifies a need to change resource provisioning or configuration, for example, the intent manager can act by utilizing orchestration and network management.

Our automation platform utilizes various distinct software components in the assurance loop of the intent-manager application. The components used to do so are referred to as agents. Agents implement various concerns of the assurance loop including:

- » Collecting measurements
- » Identifying issues with respect to intent compliance
- » Proposing healing actions and predicting their outcomes
- » Considering utility and risk in decisions
- » Executing the agreed solutions.

The agents used in our automation platform can be machine-learned models, policies or applications based on other technologies. For example, manually-designed or machine-learned policies can generate solution proposals. Digital twins, which typically include machine-learned models, can be used to safely evaluate the solution proposals and estimate their effects on network behavior. Based on this estimate, an evaluation agent applies utility functions to the current state and the predicted states. A final decision is made by maximizing utility across the available solution alternatives. In this way, the autonomous domain implements a technical solution that is optimized according to customer preference, CSP strategy and overall expected business value. The automation framework would also provide conflict resolution between intent-manager applications and non-intent-aware applications.

Intent in the RAN domain: RAN network functions

RAN network functions (NFs) are characterized by multiple control loops with real-time characteristics. A key challenge of intent-based RAN NFs is therefore calculating how to address real-time characteristics, while facing dynamically changing and potentially conflicting requirements from multiple sources.

To overcome this challenge, Ericsson proposes the introduction of intent-aware RAN, which is best

achieved by allocating intent managers in RAN NFs. This constitutes the second layer of intent within the RAN domain. These intent managers implement intent assurance loops. As such, they monitor RAN performance through an observability framework and identify discrepancies with respect to the intents.

Similar to intent managers in other domains, a RAN NF intent manager collects possible solution strategies from proposing agents, explores their effect using prediction models and makes decisions using utility information. A broad range of implementations is possible for proposing agents and prediction agents, ranging from the reuse of event-condition-action policies to machine-learned models.

If, for example, the resource-controlling features are not intent-aware and the RAN intent manager is responsible for adapting them to the current requirements, it can act by reconfiguring real-time features such as radio resource partitioning or service-adaptive link adaptation. In this way, RAN management can meet real-time requirements by employing real-time optimized features, while a potentially slower intent control loop observes if the real-time features meet intent. If not, configuration changes can be applied to the real-time features.

Alternatively, some RAN features can become intent-aware. For example, an intent-aware scheduler or intent-aware link adaptation can implement self-optimization logic that uses requirements originating from intents directly. In this scenario, the RAN NF intent manager would relay certain requirements directly to the intent-aware feature. The intent manager would ensure that the configuration of intent-aware and non-intent-aware RAN features is consistent and represent an optimal overall operation strategy.

In short, the role of a RAN NF intent manager is predominantly oversight across many real-time control features. The intent manager does not interfere directly with the operation logic, but instead verifies that the RAN features are performing according to the intent requirements. If needed, it takes action through configuration.

Conclusion

Ericsson’s intent management solutions enable a high level of autonomy, combining utility evaluation with digital twins and machine-learning-based automated exploration of system behavior. By applying these concepts to multiple architectural layers and system domains with intent as the central mechanism for requirement expression and distribution, we have designed an autonomous network that is highly self-adaptive and reliable. Our concept for multi-layer, intent-based operation enables the network to reliably detect and mitigate conflicts and keep itself optimized, while reducing the need for human assistance, including manual updates of policies and use-case-specific logic.

In each of the layers of network operation, intent constitutes a control loop between the source of intent and the system that needs to comply with the intent. Declarative requirements are sent southbound, while intent reports ensure that the originator of the requirements remains informed. The models for expressing intent as well as the application programming interfaces for communicating it are very similar across the system layers. The intent architecture is the same, while different domains mainly constitute different contexts. Within each autonomous domain, an intent manager coordinates the actions needed for complying to the intent, by maximizing the overall utility.

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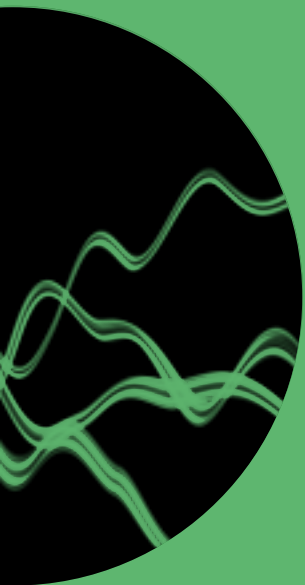


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